LIE ALGEBRA COHOMOLOGY AND SEIBERG-WITTEN THEORY (WORKING DRAFT)

AHSAN Z. KHAN

Harvard University,
Center for Mathematical Sciences and Applications,
Cambridge, MA 02138

ABSTRACT. We prove that the perturbative holomorphic-topological observables of pure $\mathcal{N}=2$ gauge theory with gauge group $SU(2)$ form the Poisson vertex algebra generated by a Virasoro element X of central charge zero and an odd primary Y of weight three, subject only to $X^2=0$, as conjectured in earlier work [K26]. Equivalently, we compute the relative Lie algebra cohomology $H^*(\mathfrak{sl}_2[z,w]/(w^2),\mathfrak{sl}_2)$. The proof views the b-ghosts as points on a line: locality pushes any discrepancy onto the small diagonal, where all points collide; a depth argument leaves it no room there; and two deformations of the double line $w^2=0$, into a parabola and into a pair of lines, eliminate what remains. As a consequence we obtain a trigraded refinement of the Schur index, a constant term identity, and, via the affine Grassmannian, the derived endomorphisms of the unit of the rank-one coherent Satake category.

1. INTRODUCTION

The holomorphic-topological twist of a four-dimensional $\mathcal{N}=2$ theory [Kap06] endows the space of local operators in the cohomology of the holomorphic-topological supercharge with the structure of a Poisson vertex algebra [OY20, BBBDN20]. In [K26] we studied this Poisson vertex algebra for the pure $\mathcal{N}=2$ gauge theory with gauge group $SU(2)$, originally studied by Seiberg and Witten [SW94]. Using the description of the twisted vector multiplet as a holomorphic-topological BF theory [ESW22] and the non-renormalization theorem of [BG25], the algebra of holomorphic-topological observables is computed, to all orders in perturbation theory, by the basic part of a classical bc-ghost system. As a graded vector space it is a relative Lie algebra cohomology,

$$\mathsf{Obs}_{\mathsf{HT}}(\mathfrak{sl}_2) = H^*\big(\mathfrak{sl}_2[z,w]/(w^2),\mathfrak{sl}_2\big). \tag{1.1}$$

Here $\mathfrak{sl}_2[z,w]/(w^2) := \mathfrak{sl}_2\otimes\mathbb{C}[z,w]/(w^2) = \mathfrak{sl}_2[z]\ltimes w\,\mathfrak{sl}_2[z]$ is the current algebra of the Takiff algebra $\mathfrak{sl}_2\otimes\mathbb{C}[w]/(w^2)$ [Tak71]. The even square-zero variable w is the parameter $\varepsilon=dz$ of [K26], and cochains are polynomial in the modes; see §2.2 for the precise conventions.

In [K26] we proposed an explicit answer. Let A be the Poisson vertex algebra generated by an even field X and an odd field Y with λ-brackets

$$\{X_\lambda X\} = \partial X + 2\lambda X, \qquad \{X_\lambda Y\} = \partial Y + 3\lambda Y, \qquad \{Y_\lambda Y\} = 0, \tag{1.2}$$

divided by the smallest Poisson vertex ideal containing X^2. Thus X is a Virasoro element of central charge zero and Y is a primary field of conformal weight three. As a differential supercommutative algebra [K26, (2.33)],

$$A = \frac{\mathbb{C}[X,\partial X,\partial^2 X,\ldots]\otimes\Lambda[Y,\partial Y,\partial^2 Y,\ldots]}{\langle X^2,\ XY,\ X\partial Y,\ Y\partial Y\rangle_\partial}, \tag{1.3}$$

Email address: ahsankhan@g.harvard.edu.
Date: October 25, 2026.

where $\langle\cdots\rangle_\partial$ denotes the differential ideal generated by the displayed elements. We constructed a homomorphism of Poisson vertex algebras

$$\varphi : A \longrightarrow \mathsf{Obs}_{\mathsf{HT}}(\mathfrak{sl}_2),$$

sending X and Y to the classes of the gaugino composites $-\delta_{ij}b^i\partial c^j$ and $\epsilon_{ijk}b^ib^jb^k$ (below we rescale the latter by $1/6$, which is immaterial; see §2.3), and conjectured that it is an isomorphism [K26, Conjecture 1]. The evidence was an equality of spin- and B-refined Euler characteristics, together with a direct computer calculation of the relative cohomology through spin 20. We also stressed in [K26, §5] that a proof would be of independent mathematical interest. The purpose of this note is to give one.

Theorem 1.1. *The map $\varphi : A \to \mathsf{Obs}_{\mathsf{HT}}(\mathfrak{sl}_2)$ is an isomorphism of Poisson vertex algebras. In particular,*

$$H^*\big(\mathfrak{sl}_2[z,w]/(w^2),\mathfrak{sl}_2\big) \;\cong\; \frac{\mathbb{C}[X,\partial X,\ldots]\otimes\Lambda[Y,\partial Y,\ldots]}{\langle X^2,\ XY,\ X\partial Y,\ Y\partial Y\rangle_\partial}$$

as differential supercommutative algebras graded by spin, ghost number and B-number.

In physical terms, every perturbative holomorphic-topological local operator of Seiberg–Witten theory is a differential polynomial in the two gaugino composites representing X and Y, and every relation among such operators is a consequence of $X^2 = 0$ and the axioms of a Poisson vertex algebra.

Theorem 1.1 determines the trigraded dimensions of the observables. The following is the Hilbert–Poincaré series of A computed in [K26, (2.65)]; see Corollary 9.1 for the dimensions sector by sector.

Corollary 1.2. *Let t, q and y keep track of the ghost number, the spin and the B-number. Then*

$$P_{\mathsf{Obs}_{\mathsf{HT}}(\mathfrak{sl}_2)}(t,q,y) = \sum_{n\geq 0} q^{n(n+1)}\,t^{2n}y^n\,\frac{(-tqy^2;q)_n}{(q;q)_n}.$$

In particular, the constant term identity for the B-refined Schur index of the pure $SU(2)$ theory, which was implied by the conjecture and checked to high order in [K26, §3.6, footnote 10], is now a theorem (Corollary 9.2).

The proof of Theorem 1.1 is conceptual. It does not use the Euler characteristic computations of [K26] or any computer calculation; the identities of characters observed in [K26] are consequences rather than ingredients. Besides the setup of [K26] and classical facts such as Chevalley's theorem on coinvariants and the Hochschild–Serre decomposition, its only external inputs are the relative Bernstein–Gelfand–Gelfand (BGG) resolution of the current algebra $\mathfrak{sl}_2[z]$, due to Garland and Lepowsky [GL76, Lep79], and the theorem of Fishel, Grojnowski and Teleman [FGT08, Theorem B] that, for an *odd* square-zero variable s, the relative cohomology of $\mathfrak{sl}_2[z,s]$ is a free algebra on explicit generators.

1.1. **The idea of the proof.** Injectivity of φ is the easier half of Theorem 1.1. In §4 we construct, for every element of the graded dual of A, an explicit Chevalley–Eilenberg cycle that detects it. The cycles live on the Cartan current algebra $\mathfrak{h}[z] \ltimes w\,\mathfrak{sl}_2[z]$, where the chain complex becomes a Koszul complex, and a change of variables from power sums to elementary symmetric functions produces them in closed form.

The difficulty lies in surjectivity: we have to show that the relative cohomology contains nothing beyond $\varphi(A)$, without having any independent description of it. Our argument is geometric, and the rest of this subsection explains its main idea.

Ghosts as points. Fix the B-number B, that is, the number of b-ghosts. The derivations T_m, $m \geq 1$, which lower the mode of every b-ghost by m and annihilate the c-ghosts, commute with the BRST differential. On the graded dual of the cohomology they act as the power sums $\sum_i t_i^m$ of B commuting variables $t_1,\ldots,t_B$, which we think of as the *positions* of the b-ghosts on the z-line. In

this way the dual of the cohomology at B-number B becomes a coherent sheaf $\mathcal{K}_B$, graded by the number p of c-ghosts, on the configuration space

$$\mathrm{Sym}^B \mathbb{A}^1 = \{\text{effective divisors } t_1 + \cdots + t_B \text{ on the } z\text{-line}\} \;\cong\; \mathbb{A}^B.$$

This is more than a reformulation: the relative BGG resolution presents $\mathcal{K}_B$ as the homology of a complex of vector bundles of length B on $\mathrm{Sym}^B \mathbb{A}^1$ (§5).

The algebra A defines a second sheaf on the same space, and here the dictionary is pleasingly literal. The generator $X \mapsto -\sum_a b^a \partial c^a$ carries a single b-ghost: it is a *simple point*. The generator $Y \mapsto b^1 b^2 b^3$ carries three b-ghosts at the same location: it is a *triple point*. Accordingly, the part of A with r factors of X and s factors of Y is dual to the pushforward of the structure sheaf along the finite map

$$\nu_{r,s} : \mathrm{Sym}^r \mathbb{A}^1 \times \mathrm{Sym}^s \mathbb{A}^1 \longrightarrow \mathrm{Sym}^{r+3s} \mathbb{A}^1, \qquad (D, D') \longmapsto D + 3D',$$

which is the normalization of the closure of the stratum of divisors with r simple and s triple points; the relations $X^2 = XY = X\partial Y = Y\partial Y = 0$ become exactly the vanishing orders along diagonals that carve out this normalization (Lemma 4.1). Injectivity of φ gives a surjection of sheaves

$$\psi_B : \mathcal{K}_B \twoheadrightarrow \mathcal{A}_B^\vee := \bigoplus_{r+3s=B} \nu_{r,s*}\mathcal{O},$$

and Theorem 1.1 asserts that its kernel, the *defect*, vanishes. Three mechanisms kill the defect, which we call *separation*, *collision* and *deformation*; see Figure 1.

Separation. Over the space of pairs (D_1, D_2) of divisors with disjoint supports, which is étale over $\mathrm{Sym}^B \mathbb{A}^1$ via $(D_1, D_2) \mapsto D_1 + D_2$, both sheaves factorize:

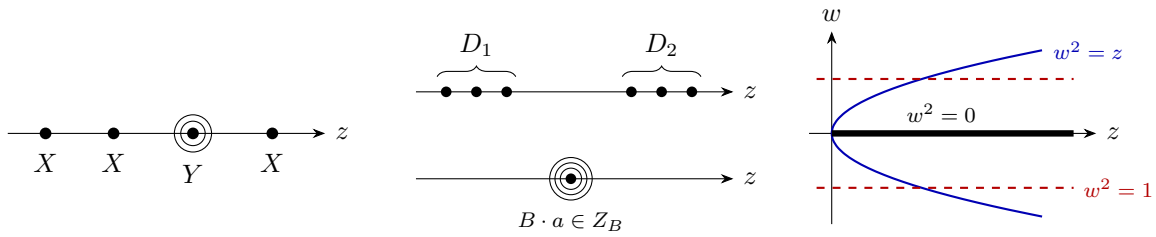
$$\mathcal{K}_B \cong \mathcal{K}_{B_1} \boxtimes \mathcal{K}_{B_2}, \qquad \mathcal{A}_B^\vee \cong \mathcal{A}_{B_1}^\vee \boxtimes \mathcal{A}_{B_2}^\vee.$$

For $\mathcal{A}^\vee$ this is the Chinese remainder theorem. For $\mathcal{K}$ it is the Lie-algebraic shadow of locality: the homology of $\mathfrak{sl}_2[z]$ with coefficients in the tensor product of two modules supported at disjoint finite sets of points, each built from evaluation modules $V_{2p}(a)$, is the tensor product of the two homologies. To prove it, we make the ghosts bosonic. If the square-zero variable w is replaced by an odd variable s, the b-ghosts commute instead of anticommuting, exterior powers become symmetric powers, the triple point Y is replaced by a double point, and the relative cohomology becomes the free algebra computed by Fishel, Grojnowski and Teleman [FGT08]. Its dual is again a sheaf on the configuration spaces of ghosts, and because the algebra is free, this sheaf visibly factorizes. Base change transfers the factorization to the fibres of the bosonic family, and every module built from evaluation modules $V_{2p}(a)$, in particular every fibre of our fermionic family, is obtained from these fibres by short exact sequences and direct summands (§6 and Appendix A). Thus the free answer for odd s certifies locality for even w. The resulting structure is reminiscent of the factorization structures of chiral algebras [BD04]. By induction on B, the map ψ_B is therefore an isomorphism away from the *small diagonal*

$$Z_B = \{B \cdot a : \; a \in \mathbb{A}^1\} \subset \mathrm{Sym}^B \mathbb{A}^1,$$

where all B ghosts collide. Hence the defect is supported on Z_B, a line of codimension $B-1$.

Collision. A sheaf supported in high codimension cannot hide in the homology of a short complex of vector bundles. This is the principle behind the acyclicity lemma of Peskine and Szpiro [PS73] (see also [BE73]), and here the numerology works out on the nose. The top term of the BGG complex, which corresponds to the pure-X sector, is a line bundle and splits off by a rank argument. What remains has length $B-1$, exactly the codimension of Z_B, while $\mathcal{A}_B^\vee$ is deep along Z_B: the preimage of Z_B under $\nu_{r,s}$ is a line in the $(r+s)$-dimensional space $\mathrm{Sym}^r \mathbb{A}^1 \times \mathrm{Sym}^s \mathbb{A}^1$, so its codimension $r+s-1$ is at least the number $p = r$ of c-ghosts as soon as $s \geq 1$. A depth count then forces the



(a) ghosts as points (b) separation and collision (c) deformation

FIGURE 1. The three mechanisms. (a) A cohomology class of B-number B is a function of the positions of B ghosts on the z-line; the generator X is a simple point and Y is a triple point. (b) When the ghosts form separated clusters the cohomology factorizes, so any defect is supported on the small diagonal Z_B, where all ghosts collide; there it is removed by a depth argument. (c) In the two lowest degrees the double line $w^2 = 0$ is deformed into the parabola $w^2 = z$ and into the pair of lines $w^2 = 1$.

defect to vanish in every degree $p \geq 2$ (§7). In short: *locality pushes every possible error onto the deepest stratum, and the BGG complex is too short to hide anything there.*

Deformation. Depth says nothing in the two lowest degrees $p = 0, 1$. Here we deform the coefficient algebra. In the (z, w)-plane, the double line $w^2 = 0$ can be opened up into the parabola $w^2 = z$, or pulled apart into the pair of lines $w^2 = 1$:

$$\mathbb{C}[z, w]/(w^2 - z) \cong \mathbb{C}[u], \qquad \mathbb{C}[z, w]/(w^2 - 1) \cong \mathbb{C}[z] \times \mathbb{C}[z].$$

For both deformed algebras the relative cohomology follows from the Garland–Lepowsky theorem, and it is tiny: $\mathbb{C}$ and $\mathbb{C} \oplus \mathbb{C}\,Y$, respectively. The corresponding spectral sequences show that an excess class in degree $p = 1$ and B-number B would have to cancel against an excess class in degree $p = 0$ and B-number $B + 2$, through a map which raises the spin by one for the parabola and by two for the pair of lines. Both cannot happen: at the lowest spin carrying an excess class, the two deformations make contradictory demands (§8). Meanwhile A itself is unaffected, as both deformations act on A with explicit contracting homotopies.

Putting the three mechanisms together proves Theorem 1.1 by induction on B (§9).

1.2. **Relation to the affine Grassmannian.** There is a second, more geometric way to think about (1.1), which places Theorem 1.1 in the context of the geometric Satake correspondence. Let G be SL_2 or PGL_2, let $\mathbf{O} = \mathbb{C}[[z]]$ and $\mathbf{K} = \mathbb{C}((z))$, and let $\mathrm{Gr}_G = G(\mathbf{K})/G(\mathbf{O})$ be the affine Grassmannian. Following a proposal of Costello reported in [CW19], the category of line operators in the holomorphic-topological twist [Kap06] of pure $\mathcal{N} = 2$ gauge theory is the derived category of $G(\mathbf{O})$-equivariant coherent sheaves on Gr_G, made monoidal by convolution, which models the collision of line operators [KS09]. With respect to the perverse coherent t-structure [AB10, BFM05], its heart is the *coherent Satake category* of Cautis and Williams [CW19]. Unlike its constructible counterpart in the geometric Satake correspondence [MV07], this category is neither semisimple nor symmetric monoidal; its simple objects correspond to Wilson–'t Hooft lines, its Grothendieck ring is the quantized K-theoretic Coulomb branch of the theory [BFM05, BFN18], and for GL_n it categorifies a quantum cluster algebra [CW19]. The construction extends to gauge theories with matter [CW23].

In any such description, bulk local operators are the derived endomorphisms of the trivial line operator, that is, of the monoidal unit, which is the skyscraper sheaf $\mathcal{O}_e$ at the base point $e = [G(\mathbf{O})]$. Niu computed this endomorphism complex [Niu22, Theorem 4.13]. The tangent space of Gr_G at e is $T_e\mathrm{Gr}_G = \mathfrak{g}(\mathbf{K})/\mathfrak{g}(\mathbf{O})$, so the self-Ext of $\mathcal{O}_e$ on the formal neighbourhood of e is

the exterior algebra $\Lambda^\bullet(\mathfrak{g}(\mathbf{K})/\mathfrak{g}(\mathbf{O}))$, and passing to derived $G(\mathbf{O})$-invariants adds the Chevalley–Eilenberg cochains of $z\mathfrak{g}(\mathbf{O})$. The result is exactly the basic bc-complex: under the residue pairing $\mathfrak{g}(\mathbf{K})/\mathfrak{g}(\mathbf{O}) \cong (\mathfrak{g}(\mathbf{O})\,dz)^\vee$, the b-ghosts become tangent vectors to Gr_G at e, and the c-ghosts become the generators of the $G(\mathbf{O})$-equivariant cohomology. Consequently Theorem 1.1 computes the derived endomorphisms of the unit of the coherent Satake category,

$$\mathrm{Ext}^\bullet(\mathcal{O}_e, \mathcal{O}_e) \;\cong\; H^*\big(\mathfrak{sl}_2[z,w]/(w^2), \mathfrak{sl}_2\big) \;\cong\; A, \tag{1.4}$$

as trigraded vector spaces (Corollary 10.1); here Ext is taken in the $G(\mathbf{O})$-equivariant category and graded by loop rotation. In this dictionary, which we spell out in §10, the B-number is the polyvector degree, the remaining degree p is the $G(\mathbf{O})$-equivariant degree, and the spin is the weight of loop rotation. The generator Y becomes the invariant trivector on the first-order normal directions $\mathfrak{g}\,z^{-1} \subset T_e\mathrm{Gr}_G$, and X becomes the class which pairs these normal directions with the first-order directions $z\mathfrak{g}$ of $G(\mathbf{O})$. The derivations T_m of §1.1 act on $T_e\mathrm{Gr}_G$ as multiplication by z^m, so the positions of the ghosts record where along the curve the corresponding tangent vectors are located. Correspondingly, the factorization at disjoint supports of §6 is the Lie-algebraic counterpart of the factorization of the Beilinson–Drinfeld Grassmannian over configurations of points [BD04, MV07], which also underlies the renormalized r-matrices of [CW19].

From this point of view Theorem 1.1 is perhaps surprising. In the constructible equivariant Satake category the derived endomorphism algebra of the unit is just $H^*_G(\mathrm{pt})$, a polynomial ring (cf. [BF08]); for coherent sheaves it is the infinitely generated algebra A, whose relations are generated by $X^2 = 0$. Niu showed that the characters of these Ext spaces reproduce the Schur index [Niu22]; Theorem 1.1 identifies the spaces themselves, with their refinement by B-number. Since the endomorphisms of the unit of a monoidal category act on all of its morphism spaces, A also acts on the spaces of junctions between line operators, such as the junctions between Wilson and 't Hooft lines studied in [Niu22], whose Euler characteristics were compared there with the defect Schur indices of [CGS16].

1.3. **Relation to other work.** The cohomology of current algebras with nilpotent coefficients has a long history. The strong Macdonald conjecture of Hanlon and Feigin [Han90, Fei88], proved by Fishel, Grojnowski and Teleman [FGT08], determines the cohomology of the truncated current algebras $\mathfrak{g}[z]/(z^N)$ and of the current superalgebra $\mathfrak{g}[z,s]$ in which s is an *odd* square-zero variable. In [K26, §5] we asked whether their techniques could be adapted to the present problem. The Lie algebra $\mathfrak{sl}_2[z,w]/(w^2)$ differs from $\mathfrak{sl}_2[z,s]$ only in the parity of the square-zero variable, but the answer is of a different nature: the cohomology rings computed in [FGT08] are free, whereas the algebra (1.3) has infinitely many relations. Accordingly, our method is different: instead of harmonic representatives, we use the module structure over symmetric functions, factorization and depth. The free answer of [FGT08] for $\mathfrak{sl}_2[z,s]$ does enter, but only through the separation step, where it certifies factorization at disjoint supports. Factorization at disjoint points of the cohomology of current algebras is also a central theme of Teleman's work on Lie algebra cohomology and the fusion rules [Tel95, Tel96]; see Remark A.3.

Organization. In §2 we recall the algebra A, the basic complex and the map φ, and reformulate Theorem 1.1 in dual form. Section 3 collects the representation-theoretic input. Section 4 proves injectivity. Section 5 sets up the sheaves on $\mathrm{Sym}^B \mathbb{A}^1$ and treats the top degree. Sections 6, 7 and 8 carry out separation, collision and deformation, respectively, and §9 assembles the proof and derives the corollaries. In §10 we explain the relation with the affine Grassmannian and the coherent Satake category in more detail, and we conclude in §11 with some remarks. Appendix A derives the factorization theorem of §6 from [FGT08, Theorem B].

Remark on AI usage. Large language models (GPT-6 Astra and Opus 5.5) were used quite extensively during this project, including in the preparation of the manuscript. Initially the author worked with Opus 5.5 to come up with a lengthy proof of Theorem 1.1 involving an Euler characteristic identity established by computer-checked telescoping certificates, and on a realization of

the invariants by charged free fermions, via the conformal embedding $\widehat{\mathfrak{so}}_3 \subset \widehat{\mathfrak{sl}}_3$ and bosonization [AGO87, Fre81]. We later simplified the proof extensively, in particular by introducing the key deformation argument of §8 to remove all dependence on fermionic vertex algebras and computer-checked certificates. GPT-6 Astra and Claude Opus 5.5 were then coordinated to produce a complete formalization of Theorem 1.1 in Lean.

Acknowledgements

This work is supported by the Center for Mathematical Sciences and Applications at Harvard University.

2. The algebra A and the basic complex

2.1. **The Poisson vertex algebra A.** Recall from [K26, §2] the free Poisson vertex algebra $V = \mathbb{C}[X, \partial X, \dots] \otimes \Lambda[Y, \partial Y, \dots]$ with λ-brackets (1.2), and the smallest Poisson vertex ideal $I \subset V$ containing X^2. Let $J \subset V$ be the differential ideal generated by X^2, XY, $X\partial Y$ and $Y\partial Y$. It was shown in [K26, §2.2] that $I = J$, so that $A = V/I$ is given by (1.3). The inclusion $J \subseteq I$ follows from

$$\{Y_\lambda X^2\} = 2X\,(2\partial Y + 3\lambda Y), \qquad \{Y_\lambda XY\} = 2\,\partial Y \cdot Y,$$

which show that XY, $X\partial Y$ and $Y\partial Y$ lie in I. The reverse inclusion says that J is a Poisson vertex ideal; it was checked directly in [K26], and it also follows from the injectivity argument of §4 (Remark 4.4).

The algebra A carries three gradings. Following [K26], X and Y have spin 2 and 3, ghost number 2 and 3, and B-number 1 and 3, and ∂ raises the spin by one. It is convenient to replace the ghost number gh by $p := \mathrm{gh} - B$, so that

$$(B, p, S)(\partial^k X) = (1, 1, k+2), \qquad (B, p, S)(\partial^k Y) = (3, 0, k+3). \tag{2.1}$$

We write S for the spin, which was denoted s in [K26], in order to free the letter s. The ideal I is homogeneous for the number of X's and the number of Y's, so $A = \bigoplus_{r,s\geq 0} A_{r,s}$, where $A_{r,s}$ is spanned by the images of monomials with r factors of the form $\partial^k X$ and s factors of the form $\partial^k Y$. By (2.1), $A_{r,s}$ lies in $(B, p) = (r + 3s, r)$. Since (B, p) determines (r, s), we write

$$A^p_B := A_{r,s} \ \text{ if } (B, p) = (r + 3s, r), \qquad A^p_B := 0 \ \text{ if } (B, p) \text{ is not of this form.} \tag{2.2}$$

2.2. **The basic complex.** As in [K26, §3.6], we identify $\mathfrak{sl}_2 \cong \mathfrak{so}_3$, with basis e_1, e_2, e_3, $[e_a, e_b] = \epsilon_{abc} e_c$, and invariant form δ_{ab}; repeated color indices are summed. In terms of the bc-system of [K26, §3.3] we use the modes

$$b^a_j := \tfrac{1}{j!}\partial^j b^a \quad (j \geq 0), \qquad c^a_n := \tfrac{1}{n!}\partial^n c^a \quad (n \geq 1), \tag{2.3}$$

all of which are odd. The basic complex of [K26, (3.67)] is the invariant part

$$\mathcal{C} := \Lambda\big(b^a_j,\, c^a_n :\ j \geq 0,\ n \geq 1,\ a = 1, 2, 3\big)^{\mathfrak{sl}_2} \tag{2.4}$$

of the exterior algebra on these modes (finite polynomial expressions only), and the BRST differential [K26, (3.65)–(3.66)] reads in modes

$$Qc^a_n = \frac{1}{2}\sum_{i=1}^{n-1} \epsilon_{abc}\, c^b_i c^c_{n-i}, \qquad Qb^a_j = \sum_{i=1}^{j} \epsilon_{abc}\, c^b_i b^c_{j-i}. \tag{2.5}$$

The gradings are $B = \#b$, $p = \#c$, and the spin $S(b^a_j) = j + 1$, $S(c^a_n) = n$; the ghost number is $\mathrm{gh} = B + p$. The differential has degree $(0, 1, 0)$ in (B, p, S), and each component of fixed (B, p, S) is finite-dimensional. We put

$$H^p_B := H^p(\mathcal{C}_B, Q), \qquad H := \bigoplus_{B,p} H^p_B = \mathsf{Obs}_{\mathsf{HT}}(\mathfrak{sl}_2). \tag{2.6}$$

Lie algebra cohomology. For a commutative $\mathbb{C}$-algebra R we write $\mathfrak{g}\otimes R$ for the corresponding current algebra, and for an ideal $(f)\subset\mathbb{C}[z,w]$ we abbreviate $\mathfrak{g}[z,w]/(f):=\mathfrak{g}\otimes\mathbb{C}[z,w]/(f)$. The algebra $\mathfrak{sl}_2[z,w]/(w^2)=\mathfrak{sl}_2[z]\ltimes w\,\mathfrak{sl}_2[z]$ has the basis of modes $e_a z^n$ and $we_a z^j$, and c_n^a and b_j^a are the coordinates dual to the modes with $n\geq 1$ and $j\geq 0$. Then Q is the Chevalley–Eilenberg differential of $\mathfrak{sl}_2[z,w]/(w^2)$, in the sign conventions of [K26], restricted to cochains that are basic for the constant subalgebra $\mathfrak{sl}_2=\mathfrak{sl}_2\otimes 1$: horizontal (no c_0^a) and invariant. We therefore write

$$H^*\big(\mathfrak{sl}_2[z,w]/(w^2),\mathfrak{sl}_2\big):=H^*(\mathcal{C},Q), \tag{2.7}$$

the cohomology of basic cochains which are polynomial in the modes. The B-number is the w-degree, and $H^p_B=H^p\big(\mathfrak{sl}_2[z],\mathfrak{sl}_2;\Lambda^B((\mathfrak{sl}_2[z]\,dz)^\vee)\big)$ with graded duals; this is the precise meaning of (1.1). Since each graded piece is finite-dimensional, this convention is the natural one, and it is part of the statement of Theorem 1.1. We use the same notation $H^*(\mathfrak{sl}_2\otimes R,\mathfrak{sl}_2)$ in §8 for $R=\mathbb{C}[z,w]/(w^2-z)$ and $R=\mathbb{C}[z,w]/(w^2-1)$, which have the same basis of modes z^n, wz^j.

Translations and positions. In modes, the holomorphic derivative of [K26] is the even derivation

$$\partial b_j^a=(j+1)\,b_{j+1}^a,\qquad \partial c_n^a=(n+1)\,c_{n+1}^a. \tag{2.8}$$

A direct computation from (2.5) gives $[Q,\partial]=-\sum_a c_1^a L_a$, where L_a is the derivation by which the constant mode e_a acts on the coordinates, $L_a b_j^c=\epsilon_{acd}b_j^d$ and $L_a c_n^c=\epsilon_{acd}c_n^d$. Hence ∂ commutes with Q on $\mathcal{C}$ and acts on H; it is the translation of the Poisson vertex algebra $\mathsf{Obs}_{\mathsf{HT}}(\mathfrak{sl}_2)$. For $m\geq 1$ we also use the even derivations

$$T_m b_j^a=b_{j-m}^a\quad(\text{zero if } j<m),\qquad T_m c_n^a=0. \tag{2.9}$$

They preserve $\mathcal{C}$ and commute with Q: both T_mQ and QT_m vanish on the c's, and on b_j^a both give $\sum_{i=1}^{j-m}\epsilon_{abc}c_i^b b_{j-m-i}^c$. The derivation T_m lowers the mode of each b-ghost by m; it is the transpose of multiplication by z^m on the coefficient module $w\,\mathfrak{sl}_2[z]$. In §5.2 it will measure the positions of the b-ghosts.

2.3. The map φ.

Put

$$\mathsf{X}:=-\sum_a b_0^a c_1^a,\qquad \mathsf{Y}:=b_0^1b_0^2b_0^3=\tfrac{1}{6}\,\epsilon_{abc}\,b_0^a b_0^b b_0^c. \tag{2.10}$$

These are the cocycles of [K26, (3.97)–(3.98)], except that Y is normalized by a factor $1/6$. Since $Y\mapsto\lambda Y$ is an automorphism of A for every $\lambda\in\mathbb{C}^\times$ (the relations and the λ-brackets are homogeneous in Y), this normalization is immaterial for Theorem 1.1, and we use it throughout.

Lemma 2.1. *The cochains* X *and* Y *are* Q*-closed, and*

$$\mathsf{X}^2=Q\big(-\epsilon_{abc}\,b_0^a b_0^b c_2^c\big),\qquad \mathsf{XY}=0,\qquad \mathsf{X}\,\partial\mathsf{Y}=Q\Big(-\tfrac{1}{4}\big(\textstyle\sum_a b_0^a b_1^a\big)^2\Big),\qquad \mathsf{Y}\,\partial\mathsf{Y}=0. \tag{2.11}$$

Proof. By (2.5), $Qb_0^a=Qc_1^a=0$. Every term of XY and of $\mathsf{Y}\,\partial\mathsf{Y}$ contains a repeated b_0^a. Using $Qc_2^c=\frac{1}{2}\epsilon_{cde}c_1^dc_1^e$ and $\epsilon_{abc}\epsilon_{cde}=\delta_{ad}\delta_{be}-\delta_{ae}\delta_{bd}$ we find $Q(\epsilon_{abc}b_0^ab_0^bc_2^c)=b_0^ab_0^bc_1^ac_1^b=-\mathsf{X}^2$; this is [K26, (3.116)], since $\partial^2 c=2c_2$. Similarly, $\sigma:=\sum_a b_0^a b_1^a$ satisfies $Q\sigma=-\epsilon_{abc}b_0^ab_0^bc_1^c$, hence $Q(\sigma^2)=2\sigma\,Q\sigma=-4\,\mathsf{Y}\sum_a b_1^ac_1^a=-4\,\mathsf{X}\,\partial\mathsf{Y}$. □

Consequently, the assignment $X\mapsto[\mathsf{X}]$, $Y\mapsto[\mathsf{Y}]$, extended multiplicatively and ∂-equivariantly, defines a homomorphism of differential algebras

$$\varphi:A\longrightarrow H.$$

Up to the normalization of Y, this is the map of [K26, §3.6]. In particular it intertwines the λ-brackets (1.2) with the Poisson vertex algebra structure of $\mathsf{Obs}_{\mathsf{HT}}(\mathfrak{sl}_2)$, which is induced from the

bc-system with $\{b^a{}_\lambda c^b\} = \delta_{ab}$ [K26, §§3.3–3.4]. Indeed, writing $b^a = b^a_0$ and $\partial c^a = c^a_1$, one finds $\{\mathsf{X}_\lambda b^a\} = (\partial+\lambda)b^a$ and $\{\mathsf{X}_\lambda \partial c^a\} = (\partial+\lambda)\partial c^a$, and the Leibniz rule gives

$$\{\mathsf{X}_\lambda \mathsf{X}\} = (\partial+2\lambda)\mathsf{X}, \qquad \{\mathsf{X}_\lambda \mathsf{Y}\} = (\partial+3\lambda)\mathsf{Y}, \qquad \{\mathsf{Y}_\lambda \mathsf{Y}\} = 0,$$

as in [K26, (3.104)–(3.113)]. Hence the map from the free algebra V preserves λ-brackets and kills X^2, so its kernel is a Poisson vertex ideal containing X^2, and therefore contains I. Lemma 2.1 makes the vanishing on the generators of I explicit. The map φ preserves the gradings (2.1), so Theorem 1.1 is equivalent to the bijectivity of

$$\varphi : A^p_B \longrightarrow H^p_B \qquad \text{for all } (B,p).$$

2.4. **Dual formulation.** For a graded vector space $W = \bigoplus_S W_S$ with finite-dimensional pieces we write $W^\vee = \bigoplus_S W_S^*$ for its graded dual. Put

$$K_{B,p} := (H^p_B)^\vee, \qquad A^\vee_{B,p} := (A^p_B)^\vee, \qquad \psi_{B,p} := \varphi^\vee : \ K_{B,p} \longrightarrow A^\vee_{B,p}. \tag{2.12}$$

Theorem 1.1 is equivalent to the statement that every $\psi_{B,p}$ is an isomorphism. We show that every $\psi_{B,p}$ is surjective in §4, and that it is injective by induction on B in §§5–9.

3. Representation-theoretic input

Throughout, $\mathfrak{l} := \mathfrak{sl}_2[z]$, V_m is the irreducible $\mathfrak{sl}_2$-module of highest weight m (of dimension $m+1$), and for $a \in \mathbb{C}$, $V_m(a)$ is the evaluation module of $\mathfrak{l}$ on which $x\, f(z)$ acts by $f(a)\, x$. For an $\mathfrak{l}$-module M we write

$$C_*(\mathfrak{l}, \mathfrak{sl}_2; M) = \big(\Lambda^\bullet(\mathfrak{l}/\mathfrak{sl}_2) \otimes M\big)_{\mathfrak{sl}_2}, \qquad C^*(\mathfrak{l}, \mathfrak{sl}_2; M) = \mathrm{Hom}_{\mathfrak{sl}_2}\big(\Lambda^\bullet(\mathfrak{l}/\mathfrak{sl}_2), M\big)$$

for the relative chains and cochains. When M is graded with finite-dimensional graded pieces, as in §2.2, we use graded duals, so that cochains are polynomial in the modes.

Theorem 3.1 (Relative BGG resolution)**.** *The trivial $\mathfrak{l}$-module has a resolution by $(\mathfrak{l}, \mathfrak{sl}_2)$-relatively projective modules*

$$\cdots \to P_2 \to P_1 \to P_0 \to \mathbb{C} \to 0, \qquad P_p = U(\mathfrak{l}) \otimes_{U(\mathfrak{sl}_2)} V_{2p}[d_p], \qquad d_p = \tfrac{p(p+1)}{2},$$

where $[d_p]$ shifts the polynomial-mode degree by d_p. Consequently, if the constant subalgebra $\mathfrak{sl}_2$ acts locally finitely on an $\mathfrak{l}$-module M, then $H_*(\mathfrak{l}, \mathfrak{sl}_2; M)$ and $H^*(\mathfrak{l}, \mathfrak{sl}_2; M)$ are computed by complexes with terms

$$(M \otimes V_{2p})^{\mathfrak{sl}_2}[d_p] \qquad \text{and} \qquad \mathrm{Hom}_{\mathfrak{sl}_2}(V_{2p}, M)[-d_p], \qquad p \geq 0,$$

respectively.

The underlying computation $H_p(z\,\mathfrak{sl}_2[z]) \cong V_{2p}[d_p]$ is the affine A_1 case of the theorem of Garland and Lepowsky [GL76], the affine analogue of Kostant's theorem [Kos61]. The resolution by generalized Verma modules, together with the resulting complex computing relative (co)homology, is due to Lepowsky [Lep79, Proposition 5.3, Corollary 5.4 and Theorem 5.5]; we use it after exchanging positive and negative currents, which reverses the sign of the mode grading. See also [Kum02]. Since $V_{2p}^{\mathfrak{sl}_2} = 0$ for $p > 0$, trivial coefficients give

$$H^*(\mathfrak{sl}_2[z], \mathfrak{sl}_2; \mathbb{C}) = \mathbb{C}, \tag{3.1}$$

concentrated in cohomological degree and polynomial-mode degree zero.

Lemma 3.2 (Hochschild–Serre)**.** *Let $\mathfrak{a} = \mathfrak{k} \ltimes \mathfrak{n}$ with $\mathfrak{k}$ semisimple, and let M be an $\mathfrak{a}$-module such that each cochain space $C^q(\mathfrak{a}; M)$ is a locally finite $\mathfrak{k}$-module. Then*

$$H^*(\mathfrak{a}; M) \cong H^*(\mathfrak{k}; \mathbb{C}) \otimes H^*(\mathfrak{a}, \mathfrak{k}; M).$$

This is proved as in [HS53]: one decomposes the cochains into $\mathfrak{k}$-isotypic components; the nontrivial components are contracted using the Casimir operator, and the trivial component contributes $H^*(\mathfrak{k};\mathbb{C})$. We only apply it to graded cochains, for which the hypothesis holds degree by degree.

The last input concerns the analogue of (2.7) in which the square-zero variable is odd. Let s be an odd variable with $s^2=0$. In the Lie superalgebra $\mathfrak{sl}_2[z,s]=\mathfrak{sl}_2[z]\ltimes s\,\mathfrak{sl}_2[z]$ the ideal $s\,\mathfrak{sl}_2[z]$ is odd and abelian, so its cochains form a symmetric rather than an exterior algebra, and

$$H^n\big(\mathfrak{sl}_2[z,s],\mathfrak{sl}_2\big)=\bigoplus_{q+k=n} H^q\big(\mathfrak{l},\mathfrak{sl}_2;\mathrm{Sym}^k(\mathfrak{l}^\vee)\big). \tag{3.2}$$

Here $\mathfrak{l}^\vee$ is the graded dual of the adjoint module $\mathfrak{l}\cong s\,\mathfrak{sl}_2[z]$, and graded duals are taken with respect to the degree in z, separately for each symmetric degree k, so that each $\mathrm{Sym}^k(\mathfrak{l}^\vee)$ has finite-dimensional graded pieces. We write $H^*(\mathfrak{l},\mathfrak{sl}_2;\mathrm{Sym}(\mathfrak{l}^\vee)):=\bigoplus_k H^*(\mathfrak{l},\mathfrak{sl}_2;\mathrm{Sym}^k(\mathfrak{l}^\vee))$. In the basis e_a of §2.2, let $\sigma_n^a\in\mathfrak{l}^\vee$, $n\ge 0$, be the coordinate dual to $e_a z^n$, regarded as an element of $\mathrm{Sym}^1(\mathfrak{l}^\vee)$, and for $n\ge 1$ let ψ_n^a be the same coordinate, regarded as a relative 1-cochain; thus ψ_n^a is the ghost c_n^a of §2.2, while the σ_n^a are commuting counterparts of the b-ghosts.

Theorem 3.3 (Fishel–Grojnowski–Teleman)**.** *The cochains*

$$\mathsf{S}_n:=\sum_{i+j=n}\sigma_i^a\,\sigma_j^a\quad(n\ge 0),\qquad \mathsf{E}_n:=\sum_{i+j=n} i\,\psi_i^a\,\sigma_j^a\quad(n\ge 1)$$

are cocycles, of cohomological degree 0 *and* 1 *and of symmetric degree* 2 *and* 1*, respectively. With graded duals, and with the product given by the cup product and the multiplication of* $\mathrm{Sym}(\mathfrak{l}^\vee)$*, the cohomology* $H^*\big(\mathfrak{l},\mathfrak{sl}_2;\mathrm{Sym}(\mathfrak{l}^\vee)\big)$ *is the free graded-commutative algebra on their classes.*

In terms of $\sigma^a(z)=\sum_{n\ge0}\sigma_n^a z^n$ and $\psi^a(z)=\sum_{n\ge1}\psi_n^a z^n$, the cocycles S_n and E_n are the coefficients of z^n in $\sigma^a(z)\,\sigma^a(z)$ and of z^{n-1} in $\frac{d\psi^a}{dz}(z)\,\sigma^a(z)$. Theorem 3.3 is [FGT08, Theorem B] for $\mathfrak{g}=\mathfrak{sl}_2$, whose invariant polynomials are generated by the invariant form, so that 1 is the only exponent; there S_n and E_n are denoted $S(-n)$ and $E(-n)$, and graded duals are called restricted duals. By (3.2) it computes the relative cohomology of $\mathfrak{sl}_2[z,s]$; see also [FGT08, Theorems 1.5 and 1.10]. In contrast with Theorem 1.1, the answer is a free algebra. Its proof in [FGT08] rests on a Laplacian identity and harmonic forms. We use Theorem 3.3 only in Appendix A, to prove the factorization theorem of §6.

4. Injectivity: separating cycles

4.1. **The dual of A.** Following [K26, §2.3], we describe the graded dual of A with the generating fields

$$X(z):=\sum_{k\ge0}\partial^k X\,\frac{z^k}{k!},\qquad Y(z):=\sum_{k\ge0}\partial^k Y\,\frac{z^k}{k!}.$$

Let $V_{r,s}\subset V$ be the span of monomials with r factors $\partial^k X$ and s factors $\partial^k Y$. For a functional η on $V_{r,s}$ put

$$\Omega_\eta(z_1,\dots,z_r;y_1,\dots,y_s):=\eta\big(X(z_1)\cdots X(z_r)\,Y(y_1)\cdots Y(y_s)\big).$$

If η belongs to the graded dual, Ω_η is a polynomial, symmetric in the z_i and alternating in the y_a, and $\eta\mapsto\Omega_\eta$ identifies the graded dual of $V_{r,s}$ with the space of such polynomials; a monomial of degree d pairs with elements of spin $2r+3s+d$. Put

$$\mathcal{O}_{r,s}:=\mathbb{C}[z_1,\dots,z_r]^{S_r}\otimes\mathbb{C}[y_1,\dots,y_s]^{S_s},\qquad F_{r,s}:=\Delta(z)^2\,\Delta(y)^3\prod_{i=1}^{r}\prod_{a=1}^{s}(z_i-y_a)^2, \tag{4.1}$$

where $\Delta(z)=\prod_{i<j}(z_i-z_j)$, and empty products are equal to one. We also write $\mathbb{C}[z,y]$ for the polynomial ring in $z_1,\dots,z_r$ and $y_1,\dots,y_s$.

Lemma 4.1. *The map $\eta \mapsto \Omega_\eta$ identifies $A_{r,s}^\vee$ with $F_{r,s}\,\mathcal{O}_{r,s}$.*

Proof. This is a reformulation of [K26, (2.54)]; we include the short argument. Since $I = J$, a functional on $V_{r,s}$ descends to $A_{r,s}$ if and only if it annihilates $J \cap V_{r,s}$, which is spanned by products of derivatives of X^2, XY, $X\partial Y$, $Y\partial Y$ with monomials. Since $X(z)^2 = e^{z\partial}(X^2)$, $X(z)Y(z) = e^{z\partial}(XY)$, $X(z)Y'(z) = e^{z\partial}(X\partial Y)$ and $Y(z)Y'(z) = e^{z\partial}(Y\partial Y)$, this happens if and only if, for all $i \neq j$ and $a \neq b$,

(a) Ω_η vanishes on $z_i = z_j$;
(b) Ω_η and $\partial_{y_a}\Omega_\eta$ vanish on $y_a = z_i$;
(c) $\partial_{y_b}\Omega_\eta$ vanishes on $y_a = y_b$.

By symmetry, (a) is equivalent to divisibility by $(z_i - z_j)^2$, and (b) is equivalent to divisibility by $(z_i - y_a)^2$. Since Ω_η is alternating, it equals $(y_a - y_b)g$ with g symmetric in y_a, y_b, and (c) says that g vanishes on $y_a = y_b$; hence (c) is equivalent to divisibility by $(y_a - y_b)^3$. The linear forms $z_i - z_j$, $z_i - y_a$, $y_a - y_b$ are pairwise non-associate primes, so (a)–(c) together say that $F_{r,s}$ divides Ω_η, and the quotient is symmetric in the z's and in the y's. □

4.2. **Cartan currents.** Choose a Cartan element h and root vectors E, F of $\mathfrak{sl}_2$, normalized here by $[h, E] = E$ and $[h, F] = -F$, let $\mathfrak{h} = \mathbb{C}h$, and consider the subalgebra and pair

$$\mathfrak{t} := \mathfrak{h}[z] \ltimes w\,\mathfrak{sl}_2[z] \;\subset\; \mathfrak{sl}_2[z,w]/(w^2), \qquad (\mathfrak{t}, \mathfrak{h}) \;\subset\; \big(\mathfrak{sl}_2[z,w]/(w^2), \mathfrak{sl}_2\big).$$

Restriction of basic cochains commutes with the differentials; dually, the inclusion of relative chains $\iota_* : C_*(\mathfrak{t}, \mathfrak{h}) \to C_*(\mathfrak{sl}_2[z,w]/(w^2), \mathfrak{sl}_2)$ is a chain map. Since $\mathcal{C}$ is the graded dual of the second complex, every cycle $\gamma \in C_*(\mathfrak{t}, \mathfrak{h})$ defines a functional $[\alpha] \mapsto \langle \alpha, \iota_*\gamma\rangle = \langle \iota^*\alpha, \gamma\rangle$ on H, which is well defined because coboundaries pair to zero with cycles.

The relative chains are $C_*(\mathfrak{t}, \mathfrak{h}) = \Lambda^\bullet(z\mathfrak{h}[z] \oplus w\,\mathfrak{sl}_2[z])_\mathfrak{h}$, the weight-zero part of the exterior algebra. Write $\xi_k := hz^k$ for $k \geq 1$. The only brackets in $\mathfrak{t}$ between the generators of this exterior algebra which do not vanish are

$$[\xi_k, wEz^m] = wEz^{m+k}, \qquad [\xi_k, wFz^m] = -wFz^{m+k}.$$

Hence the chain complex is a tensor product of three pieces: the exterior algebra on the ξ_k; the *root part* $\Lambda(wE \otimes \mathbb{C}[z]) \otimes \Lambda(wF \otimes \mathbb{C}[z])$; and the *neutral part* $\Lambda(wh \otimes \mathbb{C}[z])$, which is inert. Weight zero forces equally many factors s of each root color. Identifying wedges with alternants,

$$wEz^{m_1} \wedge \cdots \wedge wEz^{m_s} \longmapsto \det\big(u_a^{m_b}\big),$$

and similarly for F with variables $v_1, \ldots, v_s$, the root part with s factors of each color becomes $\Delta(u)\Delta(v)\,\mathcal{O}_s^{\mathrm{C}}$, where $\mathcal{O}_s^{\mathrm{C}} := \mathbb{C}[u_1, \ldots, u_s]^{S_s} \otimes \mathbb{C}[v_1, \ldots, v_s]^{S_s}$. Under this identification ξ_k acts by multiplication by

$$f_k := p_k(u) - p_k(v), \qquad p_k(u) = \textstyle\sum_a u_a^k.$$

Therefore, up to signs, the chain complex is the Koszul complex

$$\mathcal{O}_s^{\mathrm{C}} \otimes \Lambda(\xi_1, \xi_2, \ldots), \qquad d\xi_k = f_k, \tag{4.2}$$

tensored with the neutral part, on which the differential vanishes.

We now produce cycles by passing from power sums to elementary symmetric functions. Introduce a formal parameter τ, put $\mathcal{E}_u(\tau) = \prod_a (1 - u_a\tau)$, $\mathcal{E}_v(\tau) = \prod_a (1 - v_a\tau)$, and

$$L(\tau) := \log \frac{\mathcal{E}_v(\tau)}{\mathcal{E}_u(\tau)} = \sum_{k\geq 1} \frac{f_k}{k}\,\tau^k, \qquad \Xi(\tau) := \sum_{k\geq 1} \frac{\xi_k}{k}\,\tau^k,$$

$$\Theta(\tau) := \mathcal{E}_v(\tau)\,\frac{e^{-L(\tau)} - 1}{L(\tau)}\,\Xi(\tau) = \sum_{k\geq 1} \theta_k \tau^k,$$

where $(e^{-L}-1)/L$ stands for $\sum_{n\ge0}(-1)^{n+1}L^n/(n+1)!$. Since L has no constant term, each θ_k is a finite expression $\theta_k = -\xi_k/k + \sum_{j<k}\pi_{kj}\xi_j$ with $\pi_{kj}\in\mathcal{O}_s^{\mathrm{C}}$. The Koszul differential is $\mathcal{O}_s^{\mathrm{C}}$-linear with $d\Xi = L$, so

$$d\Theta = \mathcal{E}_v\big(e^{-L}-1\big) = \mathcal{E}_u - \mathcal{E}_v, \qquad \text{i.e.} \qquad d\theta_k = \begin{cases} (-1)^k\big(e_k(u)-e_k(v)\big), & k\le s,\\ 0, & k>s. \end{cases} \tag{4.3}$$

In particular $\theta_{s+1},\theta_{s+2},\dots$ are cycles. (For $s=0$ one simply has $\theta_k=-\xi_k/k$.)

Pairings. Let $\beta_x(z)=\sum_{j\ge0}(wxz^j)^\vee z^j$ for $x\in\{h,E,F\}$, and $\zeta_h(z)=\sum_{n\ge1}(hz^n)^\vee z^n$, be the generating functions of the coordinates on $\mathfrak{t}/\mathfrak{h}$ dual to the basis $\{hz^n,\ wxz^j\}$. Restricting the fields $\mathsf{X}(z) := \sum_k \partial^k\mathsf{X}\,z^k/k! = -\sum_a b^a(z)\,\frac{d}{dz}c^a(z)$ and $\mathsf{Y}(z)=b^1(z)b^2(z)b^3(z)$, where $b^a(z)=\sum_j b^a_j z^j$ and $c^a(z)=\sum_{n\ge1}c^a_n z^n$, kills the root components of c, and gives

$$\iota^*\mathsf{X}(z) = \kappa_X\,\beta_h(z)\,\zeta_h'(z), \qquad \iota^*\mathsf{Y}(y) = \kappa_Y\,\beta_h(y)\beta_E(y)\beta_F(y) \tag{4.4}$$

for nonzero constants κ_X,κ_Y. For $0\le n_1<\dots<n_r$ and $G\in\mathbb{C}[u_1,\dots,u_s]^{S_s}$ consider the chain

$$\gamma_{n,G} := \theta_{s+1+n_1}\wedge\dots\wedge\theta_{s+1+n_r}\wedge\big(\Delta(u)\Delta(v)\,G\big)\wedge\varpi, \qquad \varpi := wh\wedge whz\wedge\dots\wedge whz^{r+s-1}.$$

It is a cycle, because the Koszul differential is an $\mathcal{O}_s^{\mathrm{C}}$-linear derivation of (4.2) and each θ_{s+1+n_j} is closed by (4.3). Note that closedness holds identically in u,v; the specialization $u=v$ below is only used to evaluate pairings.

Lemma 4.2. *For a nonzero constant κ independent of n and G,*

$$\big\langle\varphi\big(X(z_1)\cdots X(z_r)\,Y(y_1)\cdots Y(y_s)\big),\ \iota_*\gamma_{n,G}\big\rangle = \pm\kappa\,F_{r,s}\,s_n(z)\,G(y), \qquad s_n(z) := \frac{\det(z_i^{n_j})}{\Delta(z)}.$$

Proof. By (4.4) the restricted cochain is a product of coordinate functions of four kinds, ζ_h', β_h, β_E and β_F, which pair only with chain factors of the corresponding kinds ξ_k, whz^j, wEz^j and wFz^j. The pairing therefore factors into three determinants. (i) The neutral factor gives $\langle\prod_i\beta_h(z_i)\prod_a\beta_h(y_a),\varpi\rangle=\pm\Delta(z_1,\dots,z_r,y_1,\dots,y_s)$, a Vandermonde determinant in all $r+s$ variables. (ii) Pairing $\prod_a\beta_E(y_a)$ with an alternant $\rho(u)$ returns $\pm\rho(y)$, and similarly for F; so a root element $\Delta(u)\Delta(v)P(u,v)$ pairs to $\pm\Delta(y)^2\,\mathrm{ev}(P)$, where $\mathrm{ev}:\mathcal{O}_s^{\mathrm{C}}\to\mathbb{C}[y]^{S_s}$ is the specialization $u=v=y$. (iii) The coefficients $\pi_{kj}\in\mathcal{O}_s^{\mathrm{C}}$ of the θ's multiply the root factor, so by (ii) they enter through ev, which is a ring homomorphism. Since $\langle\Xi(\tau),\zeta_h'(z)\rangle=\sum_k\frac{\tau^k}{k}kz^{k-1}=\tau/(1-z\tau)$, and the prefactor of Ξ in Θ specializes to $-\mathcal{E}_y(\tau)$ at $u=v=y$,

$$\mathrm{ev}\langle\theta_{s+1+n},\zeta_h'(z)\rangle = -[\tau^{s+n}]\,\frac{\mathcal{E}_y(\tau)}{1-z\tau} = -z^n\prod_{a=1}^{s}(z-y_a). \tag{4.5}$$

The Cartan factor is therefore $\det\big(\mathrm{ev}\langle\theta_{s+1+n_j},\zeta_h'(z_i)\rangle\big)=\pm\prod_{i,a}(z_i-y_a)\det(z_i^{n_j})$. Multiplying (i), (ii) and (iii), and using $\Delta(z,y)=\Delta(z)\Delta(y)\prod_{i,a}(z_i-y_a)$, gives $\pm F_{r,s}\,s_n(z)\,G(y)$ up to the product of the constants $\kappa_X^r\kappa_Y^s$ and combinatorial factors. □

Proposition 4.3. *The map φ is injective. Equivalently, every $\psi_{B,p}$ is surjective.*

Proof. As n ranges over strictly increasing sequences, the Schur polynomials $s_n(z)$ form a basis of $\mathbb{C}[z_1,\dots,z_r]^{S_r}$ [Mac95, I.3], and $G(y)$ ranges over $\mathbb{C}[y_1,\dots,y_s]^{S_s}$. By Lemma 4.2, the functionals $a\mapsto\langle\varphi(a),\iota_*\gamma_{n,G}\rangle$ on $A_{r,s}$ correspond to polynomials spanning $F_{r,s}\mathcal{O}_{r,s}$, which is all of $A_{r,s}^\vee$ by Lemma 4.1. Hence an element $a\in A_{r,s}$ with $\varphi(a)=0$ pairs to zero with all of $A_{r,s}^\vee$, so $a=0$. Different (r,s) lie in different (B,p), and φ preserves (B,p). □

Remark 4.4. The proofs of Lemma 4.1 and Proposition 4.3 only use the differential ideal J of §2.1: they show that $V/J \to H$ is injective, so that J is the kernel of $V \to H$. Since this map preserves λ-brackets, J is a Poisson vertex ideal containing X^2, hence $I \subseteq J$. Together with $J \subseteq I$, this recovers $I = J$ independently of [K26, §2.2].

5. Sheaves on configuration spaces of ghosts

5.1. **The finite BGG complex.** As an $\mathfrak{l}$-module, $w\,\mathfrak{sl}_2[z] \cong \mathfrak{sl}_2 \otimes \mathbb{C}[z]$ with $(xz^m)\cdot(wyz^n) = w[x,y]z^{m+n}$. Put

$$N_B := \Lambda^B\big(w\,\mathfrak{sl}_2[z]\big), \qquad S(wyz^j) = j+1.$$

The B-number B part $\mathcal{C}_B$ of $\mathcal{C}$ is the graded dual of the relative chain complex $C_*(\mathfrak{l}, \mathfrak{sl}_2; N_B)$, and Q is the transpose of its differential. Hence

$$K_{B,p} = H_p(\mathfrak{l}, \mathfrak{sl}_2; N_B). \tag{5.1}$$

By Theorem 3.1, $K_{B,p}$ is the homology of the complex C_B with terms

$$C_{B,p} = (N_B \otimes V_{2p})^{\mathfrak{sl}_2}[d_p], \qquad 0 \le p \le B; \tag{5.2}$$

there are no terms with $p > B$, since the highest weights occurring in N_B are at most $2B$.

Let $\mathcal{O}_B := \mathbb{C}[t_1,\dots,t_B]^{S_B}$, with $\deg t_i = 1$, be the coordinate ring of $\operatorname{Sym}^B \mathbb{A}^1$. Antisymmetrization identifies

$$N_B \cong \big(V_2^{\otimes B} \otimes \mathbb{C}[t_1,\dots,t_B] \otimes \operatorname{sgn}\big)^{S_B}, \qquad (xz^m)\cdot v = \sum_{i=1}^{B} t_i^m\, x^{(i)} v, \tag{5.3}$$

where $V_2 = \mathfrak{sl}_2$ is the adjoint representation, $x^{(i)}$ acts on the i-th tensor factor, and t_i^j records the mode z^j of the i-th factor. Multiplication by $\mathcal{O}_B$ commutes with the action of $\mathfrak{l}$.

Lemma 5.1. *Each $C_{B,p}$ is a finitely generated free graded $\mathcal{O}_B$-module, and the differentials of C_B are $\mathcal{O}_B$-linear. In particular every $K_{B,p}$ is a finitely generated graded $\mathcal{O}_B$-module.*

Proof. By Chevalley's theorem [Che55], $\mathbb{C}[t_1,\dots,t_B] \cong \mathcal{O}_B \otimes \mathcal{H}_B$ as graded $\mathcal{O}_B[S_B]$-modules, where $\mathcal{H}_B$ is a finite-dimensional graded S_B-stable complement of the ideal generated by invariants of positive degree. Since taking invariants under the finite group S_B and under $\mathfrak{sl}_2$ is exact and commutes with $\mathcal{O}_B \otimes -$, we get $C_{B,p} \cong \mathcal{O}_B \otimes \big(V_2^{\otimes B} \otimes \mathcal{H}_B \otimes \operatorname{sgn} \otimes V_{2p}\big)^{S_B \times \mathfrak{sl}_2}$, up to the grading shift. The differentials are induced by the $\mathfrak{l}$-action and the BGG maps, and therefore commute with $\mathcal{O}_B$. □

Geometrically, $\mathcal{K}_{B,p} := \widetilde{K_{B,p}}$ is the p-th homology sheaf of a complex of vector bundles of length B on $\operatorname{Sym}^B \mathbb{A}^1$, as announced in §1.1.

5.2. **Positions.** The derivation T_m of (2.9) is the transpose of the endomorphism of N_B induced by multiplication by z^m on $w\,\mathfrak{sl}_2[z]$, which by (5.3) is multiplication by the power sum $p_m(t) = \sum_i t_i^m$. Since $p_1,\dots,p_B$ generate $\mathcal{O}_B$, the $\mathcal{O}_B$-module structure of $K_{B,p}$ is the transpose of the action of the T_m on H^p_B. This is the precise sense in which the T_m measure the positions of the b-ghosts.

On the source, $T_m\,\partial^k \mathsf{X} = \frac{k!}{(k-m)!}\partial^{k-m}\mathsf{X}$ and $T_m\,\partial^k \mathsf{Y} = 3\,\frac{k!}{(k-m)!}\partial^{k-m}\mathsf{Y}$, both zero if $k < m$, as one checks from (2.10) and (2.8); on fields, $T_m \mathsf{X}(z) = z^m \mathsf{X}(z)$ and $T_m \mathsf{Y}(z) = 3z^m \mathsf{Y}(z)$. Hence T_m preserves $\varphi(A)$, and on $A^\vee_{r,s} \subset \mathbb{C}[z,y]$ the power sum $p_m(t)$ acts by multiplication by $p_m(z) + 3p_m(y)$. In other words, $\mathcal{O}_B$ acts on $A^\vee_{r,s}$ through the ring map

$$\mathcal{O}_B \longrightarrow \mathcal{O}_{r,s}, \qquad \prod_{k=1}^{B}(u - t_k) \longmapsto \prod_{i=1}^{r}(u - z_i)\prod_{a=1}^{s}(u - y_a)^3, \tag{5.4}$$

and $\psi_{B,p}$ is $\mathcal{O}_B$-linear. Finally, since $\mathcal{O}_B$ acts on N_B by $\mathfrak{l}$-module endomorphisms and the comparison of Theorem 3.1 is natural in the coefficient module, the identification $K_{B,p} \cong H_p(C_B)$ of §5.1 is $\mathcal{O}_B$-linear.

5.3. **The source sheaf.** The ring map (5.4) is the comorphism of

(5.5)
$$\nu_{r,s} : \operatorname{Sym}^r \mathbb{A}^1 \times \operatorname{Sym}^s \mathbb{A}^1 \longrightarrow \operatorname{Sym}^B \mathbb{A}^1, \qquad (\Pi_1(u), \Pi_3(u)) \longmapsto \Pi_1(u)\,\Pi_3(u)^3, \qquad B = r + 3s,$$

where we identify effective divisors with monic polynomials. The map $\nu_{r,s}$ is finite: each z_i and y_a is a root of the monic polynomial on the right of (5.4), whose coefficients lie in the image of $\mathcal{O}_B$, so $\mathbb{C}[z, y]$ and its subring $\mathcal{O}_{r,s}$ are finite over $\mathcal{O}_B$. On the dense open set where Π_1 and Π_3 are squarefree and coprime, they are recovered from $\Pi_1\Pi_3^3$ as the products over its simple and triple roots, respectively; so $\nu_{r,s}$ is birational onto its image, the closure of the stratum of divisors with r simple and s triple points. Its source is smooth, so $\nu_{r,s}$ is the normalization of this closure. Since multiplication by the nonzerodivisor $F_{r,s}$ identifies $A^\vee_{r,s}$ with $\mathcal{O}_{r,s}$ up to a shift of grading, we obtain

$$A^\vee_{B,p} \cong \begin{cases} \mathcal{O}_{r,s} = \Gamma\big(\nu_{r,s*}\mathcal{O}\big) & \text{if } (B,p) = (r+3s, r), \\ 0 & \text{otherwise,} \end{cases} \tag{5.6}$$

as ungraded $\mathcal{O}_B$-modules. (We make no claim about the uniqueness of the factorization $\Pi_1\Pi_3^3$ at the boundary of the stratum.)

5.4. **The top degree.**

Lemma 5.2. *Let $B \geq 1$. The top differential $C_{B,B} \to C_{B,B-1}$ vanishes, so that*

$$C_B = C_{B,B}[B] \oplus C'_B, \qquad C'_B \text{ concentrated in degrees } 0, \dots, B-1,$$

and $\psi_{B,B}$ is an isomorphism.

Proof. The V_{2B}-isotypic part of $V_2^{\otimes B}$ is the Cartan power, of multiplicity one and with trivial S_B-action. By (5.3), $C_{B,B}$ is therefore isomorphic to the module of alternating polynomials $\Delta(t)\mathcal{O}_B$, which is free of rank one. On the other side, $A^\vee_{B,B} = F_{B,0}\mathcal{O}_{B,0} = \Delta(z)^2\mathbb{C}[z]^{S_B}$ is free of rank one over $\mathcal{O}_B$. Since $C_{B,B}$ is the top term, $K_{B,B}$ is the kernel of the top differential, and by Proposition 4.3 it surjects onto $A^\vee_{B,B}$; so $K_{B,B}$ has rank one, like $C_{B,B}$, over the fraction field of the domain $\mathcal{O}_B$. The image of the top differential thus has rank zero; being a submodule of a free module it is torsion-free, hence zero. Now $\psi_{B,B}$ is a surjection between free modules of rank one, and its kernel, having rank zero and lying in a free module, vanishes. □

6. Separation: factorization at disjoint supports

In this section we show that $\mathcal{K}_B$ and $\mathcal{A}^\vee_B$ factorize over pairs of divisors with disjoint supports (Proposition 6.7). The argument has an algebraic and a homological part. The algebraic part is a version of the Chinese remainder theorem for clusters of roots (§6.1); it gives the factorization of $\mathcal{A}^\vee_B$ and of the coefficient modules N_B. The homological part is Theorem 6.6: the homology of $\mathfrak{l}$ turns tensor products of modules built from evaluation modules $V_{2p}(a)$ at disjoint sets of points into tensor products. We state it in §6.2 and prove it in Appendix A; this is where Theorem 3.3 enters.

6.1. **Coprime splitting.** The following lemma treats simple roots and weighted clusters of roots simultaneously, and holds over nonreduced rings.

Lemma 6.1. *Let R be a commutative ring and let $f_1, f_2 \in R[u]$ be monic of positive degrees d_1, d_2 with invertible resultant. Let $m_1, \dots, m_n$ be positive integers with $\sum_j m_j = d_1 + d_2$. Let $\mathcal{R}$ be the R-algebra generated by $x_1, \dots, x_n$ subject to the coefficient identities of*

$$\prod_{j=1}^{n} (u - x_j)^{m_j} = f_1(u) f_2(u),$$

and for $J \subseteq \{1, \dots, n\}$ let $\mathcal{R}_J$ be the R-algebra defined by $\prod_{j \in J}(u - x_j)^{m_j} = f_1(u)$ and $\prod_{j \notin J}(u - x_j)^{m_j} = f_2(u)$. Then the natural map

$$\mathcal{R} \longrightarrow \prod_{J:\ \sum_{j\in J} m_j = d_1} \mathcal{R}_J$$

is an isomorphism.

Proof. Choose $\alpha, \beta \in R[u]$ with $\alpha f_1 + \beta f_2 = 1$. In $\mathcal{R}$ we have $(f_1 f_2)(x_j) = 0$, so the elements $e_j := \beta(x_j) f_2(x_j)$ are idempotents, since $e_j(1 - e_j) = (\alpha\beta f_1 f_2)(x_j) = 0$. They split $\mathcal{R}$ into components indexed by the subsets J on which $e_j = 1$. In the component indexed by J, for $j \in J$ the element $f_2(x_j)$ is invertible and hence $f_1(x_j) = 0$, while for $j \notin J$ the element $f_1(x_j)$ is invertible and $f_2(x_j) = 0$. Put $g_1 = \prod_{j\in J}(u - x_j)^{m_j}$ and $g_2 = \prod_{j\notin J}(u - x_j)^{m_j}$. For $j \notin J$, the identity $f_1(u) - f_1(x_j) = (u - x_j) q_j(u)$ shows that $u - x_j$ is invertible modulo f_1, with inverse $-q_j(u)/f_1(x_j)$; hence g_2 is invertible modulo f_1, and $g_1 g_2 = f_1 f_2$ implies $f_1 \mid g_1$. Symmetrically $f_2 \mid g_2$. On a nonzero component, division by a monic polynomial gives $\deg g_i \geq d_i$; as $\deg g_1 + \deg g_2 = d_1 + d_2$, both are equalities, so $g_i = f_i$, and components with $\sum_{j\in J} m_j \neq d_1$ vanish. Conversely, the identities $g_i = f_i$ force the prescribed values of the idempotents. This identifies the components with the algebras $\mathcal{R}_J$. □

Fix $k_1, k_2 \geq 1$ and put $k = k_1 + k_2$. Let f_i be the universal monic polynomial of degree k_i over $\mathcal{O}_{k_i}$, and put

$$\mathcal{O}^\circ_{k_1,k_2} := (\mathcal{O}_{k_1} \otimes_{\mathbb{C}} \mathcal{O}_{k_2})\big[\mathrm{Res}(f_1, f_2)^{-1}\big], \qquad U_{k_1,k_2} := \operatorname{Spec} \mathcal{O}^\circ_{k_1,k_2}, \tag{6.1}$$

the space of pairs of divisors with disjoint supports. The map $U_{k_1,k_2} \to \mathrm{Sym}^k \mathbb{A}^1$, $(f_1, f_2) \mapsto f_1 f_2$, is étale: its tangent map $(h_1, h_2) \mapsto h_1 f_2 + f_1 h_2$, with $\deg h_i < k_i$, is an isomorphism when f_1 and f_2 are coprime. In particular $\mathcal{O}^\circ_{k_1,k_2}$ is flat over $\mathcal{O}_k$. We identify closed points of $\mathrm{Sym}^k \mathbb{A}^1$ with monic polynomials f of degree k, and write $\mathbb{C}_f$ for the residue field of $\mathcal{O}_k$ at f.

Symmetric families. Besides N_k we need its counterpart for an odd square-zero variable, in which the b-ghosts commute. For $k \geq 0$ let χ_+ and χ_- be the trivial and the sign character of S_k, and put

$$N_k^\pm := \big(V_2^{\otimes k} \otimes \mathbb{C}[t_1, \dots, t_k] \otimes \chi_\pm\big)^{S_k}, \qquad (xz^m) \cdot v = \sum_{i=1}^{k} t_i^m\, x^{(i)} v, \tag{6.2}$$

in the notation of (5.3). Thus $N_k^- = N_k \cong \Lambda^k \mathfrak{l}$, while symmetrization identifies N_k^+ with the symmetric power $\mathrm{Sym}^k \mathfrak{l}$ of the adjoint module, the *bosonic family*. In both cases $\mathcal{O}_k$ acts by $\mathfrak{l}$-module endomorphisms, the power sum $p_m(t)$ acting as the derivation of $\Lambda\mathfrak{l}$, respectively $\mathrm{Sym}\,\mathfrak{l}$, induced by multiplication by z^m on $\mathfrak{l}$. As in the proof of Lemma 5.1, $N_k^\pm \cong \mathcal{O}_k \otimes W_k^\pm$ for a finite-dimensional $\mathfrak{sl}_2$-module $W_k^\pm$, compatibly with the actions of $\mathcal{O}_k$ and $\mathfrak{sl}_2$. For $k = k_1 + k_2$, the inclusion of S_k-invariants into $S_{k_1} \times S_{k_2}$-invariants, with $\mathbb{C}[t_1, \dots, t_k] = \mathbb{C}[t_1, \dots, t_{k_1}] \otimes \mathbb{C}[t_{k_1+1}, \dots, t_k]$, defines a map of $\mathfrak{l}$-modules

$$\iota_{k_1,k_2} : N_k^\pm \longrightarrow N_{k_1}^\pm \otimes N_{k_2}^\pm, \tag{6.3}$$

which is $\mathcal{O}_k$-linear for the inclusion $\mathcal{O}_k \subset \mathcal{O}_{k_1} \otimes \mathcal{O}_{k_2}$. On $N_k^+ = \mathrm{Sym}^k \mathfrak{l}$ it is the component of bidegree (k_1, k_2) of the coproduct of $\mathrm{Sym}\,\mathfrak{l}$ for which the elements of $\mathfrak{l}$ are primitive.

Lemma 6.2. *The map ι_{k_1,k_2} induces an isomorphism of $\mathfrak{l}$-modules*

$$N_k^{\pm} \otimes_{\mathcal{O}_k} \mathcal{O}^{\circ}_{k_1,k_2} \cong \left(N_{k_1}^{\pm} \otimes_{\mathbb{C}} N_{k_2}^{\pm}\right)\left[\mathrm{Res}^{-1}\right]. \tag{6.4}$$

In particular, for coprime monic polynomials f_1, f_2 of degrees k_1, k_2 it induces an isomorphism $N_k^{\pm} \otimes_{\mathcal{O}_k} \mathbb{C}_{f_1 f_2} \cong (N_{k_1}^{\pm} \otimes_{\mathcal{O}_{k_1}} \mathbb{C}_{f_1}) \otimes (N_{k_2}^{\pm} \otimes_{\mathcal{O}_{k_2}} \mathbb{C}_{f_2})$.

Proof. Lemma 6.1, with all $m_j = 1$, decomposes $\mathbb{C}[t_1, \dots, t_k] \otimes_{\mathcal{O}_k} \mathcal{O}^{\circ}_{k_1,k_2}$ into the product, over the subsets $J \subset \{1, \dots, k\}$ with $|J| = k_1$, of the algebras of ordered roots of f_1 and of f_2, the roots of f_1 being labelled by J. The group S_k permutes these factors transitively, and the stabilizer of the factor with $J = \{1, \dots, k_1\}$ is $S_{k_1} \times S_{k_2}$. Since finite-group invariants commute with base change (averaging is a projection), projection onto this factor identifies the left-hand side of (6.4) with the $S_{k_1} \times S_{k_2}$-invariants of $V_2^{\otimes k} \otimes \chi_{\pm}$ tensored with this factor, which is the right-hand side. The composite is $\iota_{k_1,k_2} \otimes \mathcal{O}^{\circ}_{k_1,k_2}$, and the formula for the action of $\mathfrak{l}$ is the same on both sides. The second statement follows by specializing at the closed point (f_1, f_2) of U_{k_1,k_2}. □

Lemma 6.3. *Let $a \in \mathbb{C}$, $m \geq 1$, and $M := N_m^{\pm} \otimes_{\mathcal{O}_m} \mathbb{C}_{(u-a)^m}$. Then $M \cong V_2^{\otimes m}$ as an $\mathfrak{sl}_2$-module, and M has a finite filtration $M = F^0 \supset F^1 \supset \cdots$ by $\mathfrak{l}$-submodules such that $\mathfrak{l}$ acts on each F^d/F^{d+1} through evaluation at a. For N_m^+ one has $M/F^1 \cong \mathrm{Sym}^m(V_2)(a)$.*

Proof. Put $\bar{t}_i := t_i - a$. Since $\prod_i (u - t_i) = (u-a)^m$ means $\prod_i (v - \bar{t}_i) = v^m$ with $v = u - a$, the fibre of $\mathbb{C}[t_1, \dots, t_m]$ at $(u-a)^m$ is the coinvariant algebra $\mathcal{Q} := \mathbb{C}[\bar{t}_1, \dots, \bar{t}_m]/(e_1(\bar{t}), \dots, e_m(\bar{t}))$. It is graded by the degree in the $\bar{t}_i$, and by Chevalley's theorem [Che55] it is isomorphic to the regular representation of S_m, with $\mathcal{Q}^{S_m} = \mathcal{Q}^0 = \mathbb{C}$. Hence $M = (V_2^{\otimes m} \otimes \mathcal{Q} \otimes \chi_{\pm})^{S_m} \cong V_2^{\otimes m}$ as $\mathfrak{sl}_2$-modules. By (6.2), $x(z-a)^j$ acts on M by $\sum_i \bar{t}_i^{\,j} x^{(i)}$, which raises the degree in $\mathcal{Q}$ by j. So $F^d := (V_2^{\otimes m} \otimes \mathcal{Q}^{\geq d} \otimes \chi_{\pm})^{S_m}$ is an $\mathfrak{l}$-submodule, and $x(z-a)^j$ acts by zero on F^d/F^{d+1} for $j \geq 1$. Finally, for χ_+ we have $M/F^1 = (V_2^{\otimes m} \otimes \mathcal{Q}^0)^{S_m} = \mathrm{Sym}^m(V_2)$, with $\mathfrak{l}$ acting through evaluation at a. □

6.2. **Factorization of homology.** We work with relative chains. For an $\mathfrak{l}$-module M on which $\mathfrak{sl}_2$ acts locally finitely, put $\mathrm{CE}_\bullet(M) := C_*(\mathfrak{l}, \mathfrak{sl}_2; M)$. Since $\mathfrak{sl}_2$-coinvariants are exact on such modules, $\mathrm{CE}_\bullet$ is an exact functor. For two such modules, the diagonal $\mathfrak{l} \to \mathfrak{l} \oplus \mathfrak{l}$ and the Künneth isomorphism give a natural chain map

$$\Delta_{M_1,M_2} : \mathrm{CE}_\bullet(M_1 \otimes M_2) \longrightarrow \mathrm{CE}_\bullet(M_1) \otimes \mathrm{CE}_\bullet(M_2), \tag{6.5}$$

given by the shuffle coproduct of $\Lambda^\bullet(\mathfrak{l}/\mathfrak{sl}_2)$ and the identity of $M_1 \otimes M_2$. For finite-dimensional modules it is the predual of the cup product.

Definition 6.4. For a finite set $Z \subset \mathbb{C}$ and $\mathbf{p} = (p_a)_{a \in Z} \in \mathbb{Z}_{\geq 0}^Z$ put $V_{2\mathbf{p}}(Z) := \bigotimes_{a \in Z} V_{2p_a}(a)$ and $|\mathbf{p}| := \sum_a p_a$. Let $\mathcal{L}(Z)$ be the class of finite-dimensional $\mathfrak{l}$-modules admitting a finite filtration by submodules whose subquotients are isomorphic to modules $V_{2\mathbf{p}}(Z)$.

Lemma 6.5. *Let f be a monic polynomial of degree k with set of roots Z. Then $N_k^{\pm} \otimes_{\mathcal{O}_k} \mathbb{C}_f$ lies in $\mathcal{L}(Z)$.*

Proof. Write $f = \prod_{a \in Z} (u-a)^{m_a}$. Applying Lemma 6.2 repeatedly, $N_k^{\pm} \otimes \mathbb{C}_f \cong \bigotimes_{a \in Z} M^{(a)}$ with $M^{(a)} := N_{m_a}^{\pm} \otimes \mathbb{C}_{(u-a)^{m_a}}$. By Lemma 6.3, each $M^{(a)}$ has a filtration by evaluation modules at a whose underlying $\mathfrak{sl}_2$-modules are subquotients of $V_2^{\otimes m_a}$, hence direct sums of modules V_{2p}. Tensoring and refining these filtrations gives a filtration of $\bigotimes_a M^{(a)}$ with subquotients of the form $V_{2\mathbf{p}}(Z)$. □

Theorem 6.6. *Let $Z_1, Z_2 \subset \mathbb{C}$ be disjoint finite sets, and let $M_1 \in \mathcal{L}(Z_1)$ and $M_2 \in \mathcal{L}(Z_2)$. Then Δ_{M_1,M_2} is a quasi-isomorphism.*

We prove Theorem 6.6 in Appendix A, by comparison with the bosonic family. By Theorem 3.3, the homology of N_k^+ is the dual of a free algebra. As a module over $\mathcal{O}_k$, its part of degree q consists of polynomials in q variables of weight one and $(k-q)/2$ variables of weight two, alternating in the former and symmetric in the latter: the generators E_n are simple points and the S_n are double points. Since there are no relations, no vanishing conditions along diagonals appear, and factorization over U_{k_1,k_2} can be read off. Because the Chevalley–Eilenberg complexes of N_k^+ are flat over $\mathcal{O}_k$, this factorization passes to every closed fibre of N_k^+. Finally, every module in $\mathcal{L}(Z)$ is built from such fibres by short exact sequences and direct summands, and the class of pairs (M_1, M_2) for which Δ_{M_1,M_2} is a quasi-isomorphism is stable under these operations.

6.3. **Factorization of the sheaves.** Fix $B = B_1 + B_2$ with $B_1, B_2 \geq 1$, and let $\mathcal{O}^\circ_{B_1,B_2}$ and U_{B_1,B_2} be as in (6.1).

Proposition 6.7. *On U_{B_1,B_2} there are isomorphisms of $\mathcal{O}^\circ_{B_1,B_2}$-modules*

$$K_{B,p} \otimes_{\mathcal{O}_B} \mathcal{O}^\circ_{B_1,B_2} \cong \bigoplus_{p_1+p_2=p} \big(K_{B_1,p_1} \otimes_{\mathbb{C}} K_{B_2,p_2}\big)\big[\mathrm{Res}^{-1}\big], \tag{6.6}$$

$$A^\vee_{B,p} \otimes_{\mathcal{O}_B} \mathcal{O}^\circ_{B_1,B_2} \cong \bigoplus_{p_1+p_2=p} \big(A^\vee_{B_1,p_1} \otimes_{\mathbb{C}} A^\vee_{B_2,p_2}\big)\big[\mathrm{Res}^{-1}\big]. \tag{6.7}$$

We only use these as abstract isomorphisms of modules; their compatibility with ψ plays no role.

Proof. For (6.6) we first construct a comparison of perfect complexes. The augmented complex $P_\bullet \otimes P_\bullet \to \mathbb{C}$, with the diagonal action of $\mathfrak{l}$, is exact by the Künneth formula and split over $\mathfrak{sl}_2$, so relative projectivity of $P_\bullet$ provides a diagonal $P_\bullet \to P_\bullet \otimes P_\bullet$ lifting the identity of $\mathbb{C}$. On each inducing space V_{2p} it involves only finitely many terms. For $\mathfrak{l}$-modules M_1, M_2 it induces a chain map

$$(P_\bullet \otimes M_1 \otimes M_2)_{\mathfrak{l}} \longrightarrow (P_\bullet \otimes M_1)_{\mathfrak{l}} \otimes (P_\bullet \otimes M_2)_{\mathfrak{l}},$$

defined over any coefficient ring and compatible with base change. The relative Chevalley–Eilenberg resolution $U(\mathfrak{l}) \otimes_{U(\mathfrak{sl}_2)} \Lambda^\bullet(\mathfrak{l}/\mathfrak{sl}_2)$, whose diagonal is the shuffle coproduct, induces Δ_{M_1,M_2} in the same way, and comparison maps between the two resolutions identify the two chain maps up to homotopy; in particular, one is a quasi-isomorphism if and only if the other is. Since $(P_p \otimes N)_{\mathfrak{l}} \cong (N \otimes V_{2p})^{\mathfrak{sl}_2}[d_p]$, applying this to the coefficient modules of (6.4), precomposed with ι_{B_1,B_2}, produces a map of bounded complexes of finite free $\mathcal{O}^\circ_{B_1,B_2}$-modules

$$C_B \otimes_{\mathcal{O}_B} \mathcal{O}^\circ_{B_1,B_2} \longrightarrow \big(C_{B_1} \otimes_{\mathbb{C}} C_{B_2}\big)\big[\mathrm{Res}^{-1}\big]. \tag{6.8}$$

At a closed point (f_1, f_2) of U_{B_1,B_2}, its fibre is the chain map just constructed for $M_i = N_{B_i} \otimes_{\mathcal{O}_{B_i}} \mathbb{C}_{f_i}$, precomposed with the isomorphism of Lemma 6.2. By Lemma 6.5, M_i lies in $\mathcal{L}(Z_i)$, where Z_i is the set of roots of f_i, and $Z_1 \cap Z_2 = \emptyset$; so the fibre is a quasi-isomorphism by Theorem 6.6. The cone of (6.8) is a bounded complex of finite free modules whose fibers at all closed points are acyclic. Such a complex is acyclic: its homology is supported on a closed set, which, if nonempty, contains a closed point $\mathfrak{m}$, and after localizing at $\mathfrak{m}$ and cancelling isomorphisms the complex becomes a nonzero minimal complex, whose fiber at $\mathfrak{m}$ has zero differential and is nonzero. Hence (6.8) is a quasi-isomorphism. Taking homology, using flatness on the left and the Künneth formula over $\mathbb{C}$ followed by localization on the right, gives (6.6).

For (6.7), let $(B, p) = (r + 3s, r)$, since otherwise both sides vanish. By (5.4), the algebra $\mathbb{C}[z, y] \otimes_{\mathcal{O}_B} \mathcal{O}^\circ_{B_1,B_2}$ is defined by the identity

$$f_1(u) f_2(u) = \prod_{i=1}^{r} (u - z_i) \prod_{a=1}^{s} (u - y_a)^3.$$

Apply Lemma 6.1 with weight one on the z_i and weight three on the y_a, and take $S_r \times S_s$-invariants. The orbits of components are indexed by $r_1 + r_2 = r$ and $s_1 + s_2 = s$ with $r_i + 3s_i = B_i$, and the stabilizer of a component permutes each of its four lists of roots separately. By the same averaging argument,

$$\mathcal{O}_{r,s} \otimes_{\mathcal{O}_B} \mathcal{O}^{\circ}_{B_1,B_2} \cong \bigoplus_{\substack{r_1+r_2=r,\ s_1+s_2=s \\ r_i+3s_i=B_i}} \left(\mathcal{O}_{r_1,s_1} \otimes_{\mathbb{C}} \mathcal{O}_{r_2,s_2}\right)\left[\mathrm{Res}^{-1}\right].$$

By (5.6) this is (6.7). □

7. Collision: depth along the small diagonal

7.1. **The small diagonal.** For $B \geq 1$, let e_i denote the elementary symmetric functions of $t_1, \dots, t_B$, and let $\mathfrak{p}_B \subset \mathcal{O}_B$ be the ideal

$$\mathfrak{p}_B := \left(e_i - \tbinom{B}{i}(e_1/B)^i : \ 2 \leq i \leq B\right). \tag{7.1}$$

Its zero set is the small diagonal Z_B of divisors of the form $B \cdot a$, i.e. of polynomials $(u-a)^B$. Its generators are coordinates on $\mathcal{O}_B = \mathbb{C}[e_1, \dots, e_B]$ after a triangular change of variables, so they form a regular sequence of length $c = B - 1$, and $\mathfrak{p}_B$ is prime. (For $B = 1$, $\mathfrak{p}_1 = 0$.)

Depth. For a finitely generated module M over a Noetherian ring R and an ideal I, $\operatorname{depth}_I M$ is the supremum of the lengths of M-regular sequences in I, with $\operatorname{depth}_I M = \infty$ if $IM = M$ [Stacks, Definition 10.72.1, Tag 00LI]. We use: (D1) the depth lemma for short exact sequences [Stacks, Section 47.11], [BH93, §1.2]; (D2) $\operatorname{depth}_I M = \inf\{\operatorname{depth} M_{\mathfrak{q}} : \ \mathfrak{q} \supseteq I\}$ [BH93, §1.2]; (D3) $\operatorname{depth}_I M = \operatorname{depth}_{\sqrt{I}} M$, which follows from (D2); and (D4) for a finite map of Noetherian rings $R \to R'$ and a finitely generated R'-module M, $\operatorname{depth}_I M = \operatorname{depth}_{IR'} M$. To prove (D4), apply [Stacks, Lemma 10.72.11, Tag 0AUK], which concerns a local base ring, to the finite maps $R_{\mathfrak{q}} \to R'_{\mathfrak{q}}$ for primes $\mathfrak{q} \supseteq I$, whose maximal ideals are the primes of R' lying over $\mathfrak{q}$, and combine with (D2) for R and for R'.

7.2. **A variant of the acyclicity lemma.** The following is a variant of the acyclicity lemma of Peskine and Szpiro [PS73, Lemme 1.8] (see also [Stacks, Tag 00N0] and [BE73]).

Lemma 7.1. *Let R be Noetherian, let $I \subset R$ be generated by an R-regular sequence of length c, and let $C_c \to \cdots \to C_1 \to C_0$ be a complex of finitely generated free R-modules. Suppose that for $2 \leq p \leq c$ there are surjections $H_p(C) \twoheadrightarrow J_p$ whose kernels $\mathcal{M}_p$ are supported in $V(I)$, and that*

$$\operatorname{depth}_I J_p \geq p \qquad (2 \leq p \leq c).$$

Then $\mathcal{M}_p = 0$ for $2 \leq p \leq c$.

Proof. Write $\overline{C}_p := \operatorname{coker}(C_{p+1} \to C_p)$ and $\mathcal{I}_p := \operatorname{im}(C_p \to C_{p-1})$. We show by descending induction on p, from $p = c$ to $p = 2$, that $\operatorname{depth}_I \overline{C}_p \geq p - 1$ and $\mathcal{M}_p = 0$. Initially $\overline{C}_c = C_c$ is free, so $\operatorname{depth}_I \overline{C}_c \geq c$. Suppose $\operatorname{depth}_I \overline{C}_p \geq p - 1 \geq 1$. Since $\mathcal{M}_p \subseteq H_p(C) \subseteq \overline{C}_p$ is finitely generated and supported in $V(I)$, some power I^N annihilates it. Either I contains a nonzerodivisor x on $\overline{C}_p$, and then x^N kills $\mathcal{M}_p$ and acts injectively on $\overline{C}_p$; or $I\overline{C}_p = \overline{C}_p$, and then $\overline{C}_p$ vanishes at every prime containing I. In both cases $\mathcal{M}_p = 0$, so $H_p(C) = J_p$. The exact sequences

$$0 \to H_p(C) \to \overline{C}_p \to \mathcal{I}_p \to 0, \qquad 0 \to \mathcal{I}_p \to C_{p-1} \to \overline{C}_{p-1} \to 0$$

and (D1) give $\operatorname{depth}_I \mathcal{I}_p \geq \min(p-1, p-1)$ and $\operatorname{depth}_I \overline{C}_{p-1} \geq \min(c, p-2) = p - 2$, which completes the induction. □

Lemma 7.2. *For $0 < p < B$ we have $\operatorname{depth}_{\mathfrak{p}_B} A^{\vee}_{B,p} \geq p$.*

Proof. If $A^\vee_{B,p}=0$ there is nothing to prove. Otherwise $(B,p)=(r+3s,r)$ with $s\geq 1$, and by (5.6) $A^\vee_{B,p}\cong\mathcal{O}_{r,s}$ as an $\mathcal{O}_B$-module. Over a point of Z_B, the condition $\Pi_1(u)\,\Pi_3(u)^3=(u-a_0)^B$ forces $\Pi_1=(u-a_0)^r$ and $\Pi_3=(u-a_0)^s$; so the radical of $\mathfrak{p}_B\mathcal{O}_{r,s}$ is the ideal of the line

$$\big\{\big((u-a_0)^r,(u-a_0)^s\big)\,:\;a_0\in\mathbb{A}^1\big\}\subset \operatorname{Sym}^r\mathbb{A}^1\times\operatorname{Sym}^s\mathbb{A}^1\cong\mathbb{A}^{r+s}.$$

This line is the graph of a polynomial map over the coordinate $a_0=e_1(y)/s$, so its ideal is generated by $r+s-1$ of the coordinates of a polynomial coordinate system on $\mathbb{A}^{r+s}$, which form a regular sequence. By (D4) and (D3),

$$\operatorname{depth}_{\mathfrak{p}_B}A^\vee_{B,p}=\operatorname{depth}_{\mathfrak{p}_B\mathcal{O}_{r,s}}\mathcal{O}_{r,s}=\operatorname{depth}_{\sqrt{\mathfrak{p}_B\mathcal{O}_{r,s}}}\mathcal{O}_{r,s}\geq r+s-1\geq r=p.\qquad\square$$

Remark 7.3. This explains why the top term had to be split off in Lemma 5.2. For the pure-X sector $s=0$ one has $p=r=B$, but only $\operatorname{depth}_{\mathfrak{p}_B}A^\vee_{B,B}=B-1$. Removing this sector shortens the free complex to length $B-1$, which is exactly the length of the regular sequence generating $\mathfrak{p}_B$, and leaves precisely the depth bounds needed in Lemma 7.1.

8. Deformation: the two lowest degrees

8.1. **Two deformations of the square-zero algebra.** For $\ell\in\{0,1\}$ define odd derivations of the cochain algebra by

$$\delta_\ell b^a_j=0,\qquad \delta_\ell c^a_n=\frac12\sum_{\substack{i,k\geq 0\\ i+k=n-\ell}}\epsilon_{abc}\,b^b_i b^c_k, \tag{8.1}$$

with empty sums equal to zero, and put $D_\ell:=Q+\delta_\ell$. Each δ_ℓ raises B by 2 and lowers p by 1; δ_1 raises the spin by 1 and δ_0 by 2. One checks directly that $[\delta_1,\partial]=\delta_0$ and $[\delta_0,\partial]=0$.

Lemma 8.1. *(i) On $\mathcal{C}$ we have $\delta_\ell^2=0$ and $Q\delta_\ell+\delta_\ell Q=0$.*
(ii) $(\mathcal{C},D_1)$ computes $H^*(\mathfrak{sl}_2[z,w]/(w^2-z),\mathfrak{sl}_2)\cong H^*(\mathfrak{sl}_2[u],\mathfrak{sl}_2)$, via $z=u^2$ and $w=u$. The u-degree is $\mathrm{wt}_1:=2S-B$, and the cohomology is $\mathbb{C}$, in $\mathrm{wt}_1=0$.
(iii) $(\mathcal{C},D_0)$ computes $H^*(\mathfrak{sl}_2[z,w]/(w^2-1),\mathfrak{sl}_2)\cong H^*(\mathfrak{sl}_2[z]\oplus\mathfrak{sl}_2[z],\mathfrak{sl}_2^{\mathrm{diag}})$, via $w\mapsto\pm1$. The z-degree is $\mathrm{wt}_0:=S-B$, and the cohomology is $\mathbb{C}\oplus\mathbb{C}\,\mathsf{Y}$, in $\mathrm{wt}_0=0$.

Proof. The algebra $\mathbb{C}[z,w]/(w^2-z^\ell)$ has the same basis of modes z^n, wz^j as $\mathbb{C}[z,w]/(w^2)$, so $\mathcal{C}$ is also the space of basic polynomial-mode cochains of $\mathfrak{sl}_2[z,w]/(w^2-z^\ell)$ relative to $\mathfrak{sl}_2\otimes1$. The only new brackets are $[wxz^i,wyz^j]=[x,y]z^{i+j+\ell}$, whose dual is the term (8.1); hence D_ℓ is the Chevalley–Eilenberg differential of $\mathfrak{sl}_2[z,w]/(w^2-z^\ell)$ on basic cochains, in the conventions of (2.5). Since $D_\ell^2=0$ there and Q, δ_ℓ have B-degrees 0 and 2, (i) follows by separating B-degrees. (For $\ell=0$ this relies on basicness: on the full exterior algebra one finds $D_0^2=\frac12\sum_a\epsilon_{abc}\,b^b_0b^c_0\,L_a$, which vanishes on invariants.)

For $\ell=1$, the isomorphism $\mathbb{C}[z,w]/(w^2-z)\cong\mathbb{C}[u]$ sends $z^n\mapsto u^{2n}$ and $wz^j\mapsto u^{2j+1}$, so the u-degree of c^a_n and b^a_j is $2n=2S-B$ and $2j+1=2S-B$, respectively. The cohomology is given by (3.1) in the variable u.

For $\ell=0$, $\mathbb{C}[z,w]/(w^2-1)\cong\mathbb{C}[z]\times\mathbb{C}[z]$ via $w\mapsto(1,-1)$, the constants $\mathfrak{sl}_2\otimes1$ become the diagonal, and the z-degree of c^a_n and b^a_j is $n=S-B$ and $j=S-B$. By Lemma 3.2 applied to $\mathfrak{sl}_2[z]=\mathfrak{sl}_2\ltimes z\,\mathfrak{sl}_2[z]$ and by (3.1), $H^*(\mathfrak{sl}_2[z];\mathbb{C})=H^*(\mathfrak{sl}_2;\mathbb{C})$, in z-degree zero, and by the Künneth formula $H^*(\mathfrak{sl}_2[z]\oplus\mathfrak{sl}_2[z];\mathbb{C})=H^*(\mathfrak{sl}_2;\mathbb{C})^{\otimes2}$, again in z-degree zero. The homomorphism $(f,g)\mapsto(f(0),f(0))$ is a retraction onto $\mathfrak{sl}_2^{\mathrm{diag}}$, so $\mathfrak{sl}_2[z]\oplus\mathfrak{sl}_2[z]=\mathfrak{sl}_2^{\mathrm{diag}}\ltimes\mathfrak{n}$, and Lemma 3.2, which respects the z-degree, shows that the relative cohomology is two-dimensional and concentrated in z-degree zero. In $\mathrm{wt}_0=0$ the basic cochains are $\Lambda(b^1_0,b^2_0,b^3_0)^{\mathfrak{sl}_2}=\mathbb{C}\oplus\mathbb{C}\,\mathsf{Y}$, with zero differential. $\square$

8.2. **The deformations on A.** Write $X_j = \partial^j X$, $Y_j = \partial^j Y$, and similarly $\mathsf{X}_j = \partial^j \mathsf{X}$, $\mathsf{Y}_j = \partial^j \mathsf{Y}$. With $b^a(\tau) = \sum_j b^a_j \tau^j$, $c^a(\tau) = \sum_{n\ge 1} c^a_n \tau^n$ and $P_a(\tau) := \frac{1}{2}\epsilon_{abc} b^b(\tau) b^c(\tau)$, the definition (8.1) reads $\delta_1 c^a(\tau) = \tau P_a(\tau)$ and $\delta_0 c^a(\tau) = P_a(\tau) - P_a(0)$. Since $\sum_j \mathsf{X}_j \tau^j/j! = e^{\tau\partial}\mathsf{X} = -\sum_a b^a(\tau)\frac{d}{d\tau}c^a(\tau)$ and $e^{\tau\partial}\mathsf{Y} = b^1(\tau)b^2(\tau)b^3(\tau)$, and since $\sum_a b^a P_a = 3\, b^1 b^2 b^3$ and $\sum_a b^a \frac{d}{d\tau} P_a = 2\,\frac{d}{d\tau}(b^1 b^2 b^3)$, we obtain

$$\delta_1 \mathsf{X}_j = (2j+3)\,\mathsf{Y}_j, \qquad \delta_0 \mathsf{X}_j = 2\,\mathsf{Y}_{j+1}, \qquad \delta_1 \mathsf{Y}_j = \delta_0 \mathsf{Y}_j = 0. \tag{8.2}$$

Since the δ_ℓ are derivations anticommuting with Q, they act on H, and by (8.2) they preserve $\varphi(A)$. By Proposition 4.3 they are transported to derivations of A, determined by $\delta_1 X_j = (2j+3)Y_j$, $\delta_0 X_j = 2Y_{j+1}$ and $\delta_\ell Y_j = 0$.

Lemma 8.2. *Let h_1, h_0 be the odd derivations of V with*

$$h_1 X_j = h_0 X_j = 0, \qquad h_1 Y_j = X_j, \qquad h_0 Y_j = \tfrac{j}{2}\, X_{j-1} \quad (h_0 Y_0 = 0).$$

They preserve I, hence descend to A, and on A

$$\delta_1 h_1 + h_1 \delta_1 = \mathrm{wt}_1 = 2S - B, \qquad \delta_0 h_0 + h_0 \delta_0 = \mathrm{wt}_0 = S - B. \tag{8.3}$$

Proof. The derivation h_1 commutes with ∂ and sends the four generators X^2, XY, $X\partial Y$, $Y\partial Y$ of I to 0, X^2, $\frac{1}{2}\partial(X^2)$ and $2X\partial Y - \partial(XY)$, respectively; so it preserves the differential ideal I. The derivation h_0 sends the same generators to 0, 0, $\frac{1}{2}X^2$ and $-\frac{1}{2}XY$, and satisfies $[h_0, \partial] = \frac{1}{2}h_1$. Hence $h_0(\partial^k \rho) = \partial h_0(\partial^{k-1}\rho) + \frac{1}{2}h_1(\partial^{k-1}\rho) \in I$ for each generator ρ by induction on k, and h_0 preserves I. Both sides of each identity in (8.3) are even derivations, so it suffices to check them on generators, where they follow from (2.1) and the formulas for δ_ℓ: for instance $(\delta_1 h_1 + h_1\delta_1)X_j = (2j+3)X_j$ and $(\delta_0 h_0 + h_0\delta_0)Y_j = jY_j$. □

8.3. **The lowest two degrees.** Regard A as a subspace of H via φ, and put $\overline{H}^p_B := H^p_B/\varphi(A^p_B)$. No algebra structure on $\overline{H}$ is needed.

Proposition 8.3. *If $\overline{H}^p_b = 0$ for all $b < B$ and all p, then $\overline{H}^0_B = \overline{H}^1_B = 0$.*

Proof. The parabola. Fix $m > 0$ and consider the subcomplex of $(\mathcal{C}, D_1)$ with $\mathrm{wt}_1 = m$. It is finite-dimensional, since every generator has positive wt_1, and it is acyclic by Lemma 8.1(ii). As $\mathrm{wt}_1 = 2S - B$, the parity ε of B is fixed. Filter by B, i.e. by $(B-\varepsilon)/2$. The resulting spectral sequence has $E_1 = H$ (at $\mathrm{wt}_1 = m$), first differential $d_1 = \delta_1$, higher differentials

$$d_r : (B, p, S) \longmapsto (B + 2r,\ p + 1 - 2r,\ S + r),$$

and converges to zero. By (8.2), $0 \to \varphi(A) \to H \to \overline{H} \to 0$ is a short exact sequence of complexes for d_1, and $\varphi(A)$ is d_1-acyclic at $\mathrm{wt}_1 = m > 0$ by Lemma 8.2. Hence $E_2 = H(\overline{H}, \delta_1)$, and by hypothesis E_2 vanishes in all B-numbers below B.

At $(B, 0)$ the incoming d_1 starts at $\overline{H}^1_{B-2} = 0$ and the outgoing d_1 lands in $p = -1$, so $E_2 = \overline{H}^0_B$. For $r \ge 2$, outgoing differentials land in negative p and incoming ones start at B-numbers below B. So $\overline{H}^0_B$ survives to $E_\infty = 0$. Since this holds for every $m > 0$, and $\mathrm{wt}_1 = 0$ contains only the constants, $\overline{H}^0_B = 0$.

Put $\mathcal{V}_S := (\overline{H}^1_B)_S$ and $\mathcal{W}_S := (\overline{H}^0_{B+2})_S$. At the positions $(B, 1, S)$ and $(B+2, 0, S+1)$, which have the same wt_1, the E_2-terms are the kernel and the cokernel of $\delta_1 : \mathcal{V}_S \to \mathcal{W}_{S+1}$. For $r \ge 2$, all outgoing differentials land in negative p, and all incoming ones start at B-numbers at most $B - 2$. Both entries therefore survive to $E_\infty = 0$, and

$$\delta_1 : \mathcal{V}_S \xrightarrow{\ \sim\ } \mathcal{W}_{S+1} \qquad \text{for all } S. \tag{8.4}$$

(When $2S - B \le 0$ both sides vanish.) Note that no assumption about B-number $B + 2$ has been made.

The pair of lines. Now fix $m > 0$ and use D_0 in $\mathrm{wt}_0 = S - B = m$. This subcomplex is again finite-dimensional, because the only generators with $\mathrm{wt}_0 = 0$ are the three odd elements b^a_0, and

it is acyclic by Lemma 8.1(iii). Unlike wt_1, the weight wt_0 does not fix the parity of B; but D_0 preserves it, since Q preserves B and δ_0 raises it by 2. We therefore split the subcomplex according to the parity ε of B and filter each part by $(B-\varepsilon)/2$. This gives a spectral sequence with $d_1=\delta_0$ and

$$d_r : (B,p,S) \longmapsto (B+2r,\, p+1-2r,\, S+2r),$$

and Lemma 8.2 again identifies E_2 with $H(\overline{H},\delta_0)$. The argument above gives

$$\delta_0 : \mathcal{V}_S \xrightarrow{\sim} \mathcal{W}_{S+2} \qquad \text{for all } S \tag{8.5}$$

in $\mathrm{wt}_0>0$. In $\mathrm{wt}_0=0$ the whole complex is $\mathbb{C}\oplus\mathbb{C}\,\mathsf{Y}\subset\varphi(A)$ by Lemma 8.1(iii), so $\overline{H}$ vanishes there, and there are no cochains with $\mathrm{wt}_0<0$. Thus (8.5) holds for all S.

Conclusion. Suppose $\overline{H}^1_B\neq 0$, and let S be the smallest spin with $\mathcal{V}_S\neq 0$. By (8.4), $\mathcal{W}_{S+1}\cong\mathcal{V}_S\neq 0$. By (8.5), $\mathcal{W}_{S+1}\cong\mathcal{V}_{S-1}=0$. This contradiction shows $\overline{H}^1_B=0$. All filtrations used are finite in each fixed wt_ℓ, so no convergence issues arise. □

Remark 8.4. Either deformation alone would allow a pair of excess classes in $(B,1)$ and $(B+2,0)$, cancelling each other in the spectral sequence. What rules them out is that the parabola and the pair of lines demand different spin shifts, $+1$ and $+2$, for the same pair, which is impossible for a grading bounded below. This replaces any Euler characteristic or residue computation in low degrees.

9. Proof of the main theorem

Proof of Theorem 1.1. By Proposition 4.3, φ is injective and every $\psi_{B,p}$ is surjective. We show that $\psi_{B,p}$ is an isomorphism for all p, by induction on B. For $B=0$, (3.1) gives $H^*_0=\mathbb{C}=A^0_0$. Let $B\geq 1$ and assume the claim for all smaller B-numbers; equivalently, $\overline{H}^p_b=0$ for $b<B$. Write $\mathcal{M}_p:=\ker\psi_{B,p}$.

Degrees $p=0,1$ and $p=B$. Proposition 8.3 gives $\overline{H}^0_B=\overline{H}^1_B=0$, i.e. $\mathcal{M}_0=\mathcal{M}_1=0$, and Lemma 5.2 gives $\mathcal{M}_B=0$.

Separation. Let $B=B_1+B_2$ with $B_i\geq 1$. By Proposition 6.7 and the induction hypothesis, $K_{B,p}\otimes_{\mathcal{O}_B}\mathcal{O}^\circ_{B_1,B_2}$ and $A^\vee_{B,p}\otimes_{\mathcal{O}_B}\mathcal{O}^\circ_{B_1,B_2}$ are isomorphic finitely generated modules over the Noetherian ring $\mathcal{O}^\circ_{B_1,B_2}$. The base change of $\psi_{B,p}$ is a surjection between them; composing it with an abstract isomorphism in the opposite direction gives a surjective endomorphism of a finitely generated module, which is injective [Mat86, Theorem 2.4], [Stacks, Tag 05G8]. Hence the base change of $\psi_{B,p}$ is an isomorphism. By flatness, $\mathcal{M}_p\otimes_{\mathcal{O}_B}\mathcal{O}^\circ_{B_1,B_2}=0$. Now let $\mathfrak{m}\subset\mathcal{O}_B$ be a maximal ideal not containing $\mathfrak{p}_B$, i.e. a monic polynomial of degree B with at least two distinct roots. Factoring it as f_1f_2 with f_1,f_2 coprime of positive degrees gives a point of some U_{B_1,B_2} over $\mathfrak{m}$, at which the local map is flat and local, hence faithfully flat; so $(\mathcal{M}_p)_{\mathfrak{m}}=0$. Since $\mathcal{O}_B$ is a Jacobson ring, every nonempty locally closed subset of $\operatorname{Spec}\mathcal{O}_B$ contains a maximal ideal; applying this to $\operatorname{Supp}\mathcal{M}_p\setminus V(\mathfrak{p}_B)$, we conclude

$$\operatorname{Supp}_{\mathcal{O}_B}\mathcal{M}_p\subseteq V(\mathfrak{p}_B)=Z_B.$$

Collision. By Lemma 5.2, $K_{B,p}=H_p(C'_B)$ for $0\leq p\leq B-1$, where C'_B is a complex of finitely generated free $\mathcal{O}_B$-modules in degrees $0,\dots,B-1$. Apply Lemma 7.1 with $R=\mathcal{O}_B$, $I=\mathfrak{p}_B$, $c=B-1$, $J_p=A^\vee_{B,p}$ and $C=C'_B$; its depth hypothesis is Lemma 7.2. Hence $\mathcal{M}_p=0$ for $2\leq p\leq B-1$. (For $B\leq 2$ all degrees were already covered by the previous step.)

Thus every $\psi_{B,p}$ is an isomorphism, which completes the induction. Dualizing each finite-dimensional graded piece shows that φ is bijective. As explained in §2.3, φ is compatible with products, with ∂ and with the λ-brackets, so it is an isomorphism of Poisson vertex algebras. □

Corollary 9.1. *For $r, s \geq 0$,*

$$\sum_S \dim\bigl(H^r_{r+3s}\bigr)_S\, q^S = \frac{q^{(r+s)(r+s+1)+s(s+1)/2}}{(q;q)_r\,(q;q)_s},$$

and $H^p_B = 0$ if (B,p) is not of the form $(r+3s, r)$. Equivalently, Corollary 1.2 holds.

Proof. By Theorem 1.1 and Lemma 4.1, the graded dual of H^r_{r+3s} is $F_{r,s}\mathcal{O}_{r,s}$. The polynomial ring $\mathcal{O}_{r,s}$ has generators of degrees $1,\dots,r$ and $1,\dots,s$, the base spin is $2r+3s$, and $\deg F_{r,s} = r(r-1) + \frac{3}{2}s(s-1) + 2rs$; the sum of the last two is $(r+s)(r+s+1) + s(s+1)/2$. For the second statement, the ghost number of H^r_{r+3s} is $2r+3s$; setting $n = r+s$ and using the q-binomial theorem $(-x;q)_n = \sum_s \binom{n}{s}_q q^{s(s-1)/2} x^s$ with $x = tqy^2$ gives the formula of Corollary 1.2, which is [K26, (2.65)]. □

Corollary 9.2. *The constant term identity of* [K26, §3.6, footnote 10] *holds:*

$$-\frac{1}{2}\oint_{|z|=1} \frac{dz}{2\pi i z}\,(z-z^{-1})^2\,(q;q)_\infty (qz^2;q)_\infty (qz^{-2};q)_\infty (qy;q)_\infty (qyz^2;q)_\infty (qyz^{-2};q)_\infty$$
$$= \sum_{n\geq 0} q^{n(n+1)} y^n \frac{(y^2q;q)_n}{(q;q)_n}.$$

Proof. The left-hand side is the B-refined Euler characteristic $\sum(-1)^{\mathrm{gh}} \dim \mathcal{C}^{S,\mathrm{gh},B} q^S y^B$ of the basic complex, computed by the Molien integral in [K26, (3.126)]. By the Euler–Poincaré principle, applied at each fixed spin and B-number, it equals $P_{\mathsf{Obs}_{\mathsf{HT}}(\mathfrak{sl}_2)}(-1,q,y)$, which by Corollary 1.2 is the right-hand side. □

10. The affine Grassmannian and the coherent Satake category

In this section we spell out the relation with the affine Grassmannian sketched in §1.2. Let G be a connected reductive group with Lie algebra $\mathfrak{g}$, let $\mathbf{O} = \mathbb{C}[[z]]$, $\mathbf{K} = \mathbb{C}((z))$, and let $\mathrm{Gr}_G = G(\mathbf{K})/G(\mathbf{O})$, with base point $e = [G(\mathbf{O})]$. We use the model of line operators of Niu [Niu22, §4]: $\widetilde{G}(\mathbf{O})$-equivariant ind-coherent sheaves on Gr_G, where $\widetilde{G}(\mathbf{O}) = G(\mathbf{O}) \rtimes \mathbb{C}^\times$ and $\mathbb{C}^\times$ acts through a cover of loop rotation, with morphism complexes valued in graded vector spaces, i.e. in representations of $\mathbb{C}^\times$. The coherent Satake category of [CW19] is the heart of the perverse coherent t-structure on the corresponding category of coherent sheaves; in both settings the monoidal unit is the structure sheaf $\mathcal{O}_e$ of the point orbit $\mathrm{Gr}^0_G = \{e\}$ [CW19, §2].

10.1. **The unit.** Niu's computation [Niu22, Theorem 4.13] identifies the derived endomorphisms of $\mathcal{O}_e$, as a complex of $\mathbb{C}^\times$-representations, with

$$\Bigl[\mathrm{Sym}^\bullet\bigl((\mathfrak{g}(\mathbf{K})/\mathfrak{g}(\mathbf{O}) \oplus (z\mathfrak{g}(\mathbf{O}))^\vee\bigr)[-1]\bigr)\Bigr]^G \tag{10.1}$$

equipped with the Chevalley–Eilenberg differential: the first summand is the tangent space $T_e\mathrm{Gr}_G$, contributing the exterior algebra of polyvectors at e, and the second contributes the cochains computing derived invariants of the pro-unipotent group $\ker(G(\mathbf{O}) \to G)$. As noted in [Niu22, §3], this is the basic bc-complex of the vector multiplet. Explicitly, for $\mathfrak{g} = \mathfrak{sl}_2$ the residue pairing

$$\bigl\langle xz^{-j-1},\, yz^k\,dz \bigr\rangle = (x,y)\,\delta_{jk}$$

identifies $T_e\mathrm{Gr}_G = \mathfrak{g}(\mathbf{K})/\mathfrak{g}(\mathbf{O})$ with the graded dual of $\mathfrak{g}(\mathbf{O})\,dz$, and the dictionary is as follows.

basic complex and A	affine Grassmannian
b_j^a	tangent vector $e_a z^{-j-1} \in T_e\mathrm{Gr}_G = \mathfrak{g}(\mathbf{K})/\mathfrak{g}(\mathbf{O})$
c_n^a	equivariant generator $(e_a z^n)^\vee \in (z\mathfrak{g}(\mathbf{O}))^\vee$
B-number	polyvector degree
p	$G(\mathbf{O})$-equivariant degree
ghost number $\mathrm{gh} = B + p$	Ext degree
spin S	weight of loop rotation
T_m	multiplication by z^m on $T_e\mathrm{Gr}_G$
∂	infinitesimal translation of the point on the curve
Y	the invariant trivector on $\mathfrak{g}\, z^{-1} \subset T_e\mathrm{Gr}_G$
X	the pairing of $\mathfrak{g}\, z^{-1} \subset T_e\mathrm{Gr}_G$ with $z\mathfrak{g}/z^2\mathfrak{g}$

For instance $\mathsf{Y} = b_0^1 b_0^2 b_0^3$ corresponds to $e_1 z^{-1} \wedge e_2 z^{-1} \wedge e_3 z^{-1}$. The subspace $\mathfrak{g}\, z^{-1} \subset T_e\mathrm{Gr}_G$ is annihilated by $z\mathfrak{g}(\mathbf{O})$, since $[xz^n, yz^{-1}] \in \mathfrak{g}(\mathbf{O})$ for $n \geq 1$, so this trivector is invariant under $G(\mathbf{O})$; it is the volume form of the first-order normal directions. Likewise $\mathsf{X} = -\sum_a b_0^a c_1^a$ is the cocycle $z\mathfrak{g}(\mathbf{O}) \to T_e\mathrm{Gr}_G$ sending xz to $-xz^{-1}$ and the higher modes to zero.

Corollary 10.1. *Let $G = SL_2$ or PGL_2. For every n there is an isomorphism, compatible with loop rotation and spin,*

$$\mathrm{Ext}^n_{\widetilde{G}(\mathbf{O})}(\mathcal{O}_e, \mathcal{O}_e) \;\cong\; \bigoplus_{B+p=n} H^p_B \;\cong\; \bigoplus_{B+p=n} A^p_B,$$

under which the polyvector degree of (10.1) corresponds to B. In particular, the Poincaré series of $\mathrm{Ext}^\bullet(\mathcal{O}_e, \mathcal{O}_e)$, refined by polyvector degree, is the series of Corollary 1.2.

Proof. For connected G the invariants in (10.1) are the $\mathfrak{g}$-invariants, so the complex depends on G only through $\mathfrak{g} = \mathfrak{sl}_2$, and under the residue pairing it is the complex $\mathcal{C}$ of (2.4), with the polyvector degree equal to B. The first isomorphism is therefore [Niu22, Theorem 4.13], and the second is Theorem 1.1. □

The quasi-isomorphism of [Niu22] is stated for complexes. One expects it to intertwine the Yoneda product with the product of the basic complex, in which case Corollary 10.1 is an isomorphism of algebras; one also expects the E_2-structure on the endomorphisms of the unit to induce the Poisson vertex algebra structure of $\mathsf{Obs}_{\mathsf{HT}}(\mathfrak{sl}_2)$, with the λ-bracket arising as the secondary product of [BBBDN20]. We do not pursue these comparisons here.

10.2. **Positions and the Beilinson–Drinfeld Grassmannian.** In the formal setting of (10.1) all tangent vectors sit at the point $z = 0$: the operators T_m, i.e. multiplication by z^m on the torsion $\mathbf{O}$-module $T_e\mathrm{Gr}_G$, act locally nilpotently. Passing to graded duals, as in §5, spreads this out over the configuration spaces $\mathrm{Sym}^B \mathbb{A}^1$. The derived fiber of $\mathcal{K}_B$ at a divisor $t_1 + \cdots + t_B$ with distinct points computes the homology of $\mathfrak{l}$ with coefficients in $\bigotimes_i V_2(t_i)$, and the dual of $V_2(t)$ is the space $\mathfrak{g}\,(z-t)^{-1}$ of tangent vectors with a simple pole at t: under the residue pairing at t, the vector $y\,(z-t)^{-1}$ pairs with $x\,f(z)\,dz$ to give $(x,y)f(t)$. Thus the positions of §1.1 are the locations of the poles of tangent vectors to the global Grassmannian of the curve $\mathbb{A}^1$, exactly as in the Beilinson–Drinfeld Grassmannian, whose fibers over disjoint points are products of affine Grassmannians [BD04, MV07]. Theorem 6.6 and Proposition 6.7 are the Lie-algebraic form of this factorization. In the coherent Satake category, the same factorization is the origin of the renormalized r-matrices of [CW19], which arise from the Beilinson–Drinfeld Grassmannian over pairs of points. It would be interesting to identify the sheaves $\mathcal{K}_B$ with Ext sheaves along the unit section of a Beilinson–Drinfeld Grassmannian over $\mathrm{Sym}^B \mathbb{A}^1$.

10.3. **Contrast with the constructible case.** In the constructible equivariant Satake category the derived endomorphisms of the unit are $H^*_{G(\mathbf{O})}(\mathrm{pt}) = H^*_G(\mathrm{pt})$, a polynomial ring; the derived Satake equivalence describes them in terms of invariant functions on the Langlands dual Lie algebra [BF08]. For coherent sheaves the answer, for $G = SL_2$ or PGL_2, is the infinitely generated algebra A of Theorem 1.1, whose relations are generated, in the sense of Poisson vertex ideals, by $X^2 = 0$. Its Euler characteristic is the Schur index [Niu22, K26], while its full trigraded character is the refinement $P_A(t, q, y)$.

11. Concluding remarks

We close with a few comments.

- The mechanisms of separation and collision are not specific to $\mathfrak{sl}_2$. For a simple Lie algebra $\mathfrak{g}$, the dual of $H^*(\mathfrak{g}[z,w]/(w^2), \mathfrak{g})$ at fixed B-number is again the homology of a finite BGG complex of vector bundles on $\mathrm{Sym}^B \mathbb{A}^1$, and [FGT08, Theorem B] provides the bosonic family of Appendix A for every reductive $\mathfrak{g}$, now with points of weights m_i and $m_i + 1$ for the exponents m_i of $\mathfrak{g}$; we expect the separation step to extend. What is special about $\mathfrak{sl}_2$ is the explicit answer (1.3), the depth estimate of Lemma 7.2 for the corresponding sheaves, and the low-degree argument of §8, which uses the explicit action of the two deformations on A. It would be very interesting to find the analogue of A for higher rank, and to see whether a version of the deformation argument persists.
- In [K26, §4] we introduced a differential Q_{inst} on A, which we hypothesized to capture non-perturbative corrections, and translated it to the vector multiplet fields [K26, (4.40)] assuming that φ is invertible. By Theorem 1.1 this translation is now unconditional, and the cohomology of Q_{inst} computed in [K26, §4] is a statement about $\mathsf{Obs}_{\mathsf{HT}}(\mathfrak{sl}_2)$ itself.
- Since the endomorphisms of the unit act on all morphism spaces of a monoidal category, Corollary 10.1 makes every space of junctions between line operators a module over A. For PGL_2, the simplest examples are the morphism spaces involving the structure sheaf of the minuscule orbit $\mathrm{Gr}^{\omega}_G \cong \mathbb{P}^1$, which Niu studied and compared with the defect Schur indices of [CGS16]; Niu also noted that the module structure of junction spaces over the bulk local operators had not been described [Niu22, §5]. It would be interesting to describe these modules explicitly, and to relate the resulting structures to the cluster theory of the coherent Satake category [CW19, CW23].

Appendix A. Proof of the factorization theorem

In this appendix we prove Theorem 6.6, keeping the notation of §6; in particular we use Lemmas 6.2 and 6.3 for the bosonic family N_k^+. By Theorem 3.1 we have $H_*(\mathfrak{l}, \mathfrak{sl}_2; \mathbb{C}) = \mathbb{C}$, so that the augmentation $\mathrm{CE}_\bullet(\mathbb{C}) \to \mathbb{C}$ is a quasi-isomorphism. Since the shuffle coproduct is counital, it follows that

$$\Delta_{\mathbb{C},M} \quad \text{and} \quad \Delta_{M,\mathbb{C}} \quad \text{are quasi-isomorphisms for every } M. \tag{A.1}$$

A.1. **The bosonic family.** For a monic polynomial f of degree k put $\mathfrak{M}_f := N_k^+ \otimes_{\mathcal{O}_k} \mathbb{C}_f$, so that $\mathfrak{M}_1 = \mathbb{C}$. We identify $\mathrm{Sym}^k(\mathfrak{l}^\vee)$ with the graded dual of $\mathrm{Sym}^k \mathfrak{l} = N_k^+$ through the pairing $\langle \phi_1 \cdots \phi_k, x_1 \cdots x_k \rangle = \sum_{\sigma \in S_k} \prod_i \phi_i(x_{\sigma(i)})$, under which multiplication is dual to the coproduct. For $q, r \geq 0$ let

$$\mathcal{P}_{q,r} \subset \mathbb{C}[z_1, \dots, z_q, y_1, \dots, y_r] \tag{A.2}$$

be the space of polynomials which are alternating in the z_i and symmetric in the y_a, regarded as a module over $\mathcal{O}_{q+2r}$ through the ring map $\prod_j (u - t_j) \mapsto \prod_i (u - z_i) \prod_a (u - y_a)^2$. In the language of §1.1, the generators E_n of Theorem 3.3 are simple points and the S_n are double points: for commuting ghosts the invariant of lowest degree is the quadratic form, where for anticommuting ghosts it is the cubic form Y.

Lemma A.1. *For $k, q \geq 0$, there is an isomorphism of $\mathcal{O}_k$-modules $H_q(\mathfrak{l}, \mathfrak{sl}_2; N_k^+) \cong \mathcal{P}_{q,r}$ if $k = q + 2r$ with $r \geq 0$, and $H_q(\mathfrak{l}, \mathfrak{sl}_2; N_k^+) = 0$ otherwise. Under these isomorphisms, for $k = k_1 + k_2$, the map induced on homology by*

$$\Delta_{N_{k_1}^+, N_{k_2}^+} \circ \mathrm{CE}_\bullet(\iota_{k_1,k_2}) : \mathrm{CE}_\bullet(N_k^+) \longrightarrow \mathrm{CE}_\bullet(N_{k_1}^+) \otimes \mathrm{CE}_\bullet(N_{k_2}^+)$$

is, up to nonzero factors on the summands, the direct sum over $q_1 + q_2 = q$ and $r_1 + r_2 = r$ with $q_i + 2r_i = k_i$ of the inclusions $\mathcal{P}_{q,r} \subset \mathcal{P}_{q_1,r_1} \otimes \mathcal{P}_{q_2,r_2}$ of subspaces of $\mathbb{C}[z, y]$, where the first factor uses the variables $z_1, \dots, z_{q_1}$ and $y_1, \dots, y_{r_1}$, and the second factor the remaining ones.

Proof. For the grading by the degree in z, each graded piece of $\mathrm{CE}_\bullet(N_k^+)$ is finite-dimensional, so $H_q(\mathfrak{l}, \mathfrak{sl}_2; N_k^+)$ is the graded dual of $H^q(\mathfrak{l}, \mathfrak{sl}_2; \mathrm{Sym}^k(\mathfrak{l}^\vee))$. By Theorem 3.3, the latter has the basis of monomials $\mathsf{E}_{n_1} \cdots \mathsf{E}_{n_q} \mathsf{S}_{j_1} \cdots \mathsf{S}_{j_r}$ with $n_1 > \cdots > n_q \geq 1$, $j_1 \geq \cdots \geq j_r \geq 0$ and $q + 2r = k$. To a functional η on it we assign the polynomial

$$G_\eta(z; y) := \sum_{n,j} \eta\big(\mathsf{E}_{n_1} \cdots \mathsf{E}_{n_q} \mathsf{S}_{j_1} \cdots \mathsf{S}_{j_r}\big)\, z_1^{n_1 - 1} \cdots z_q^{n_q - 1}\, y_1^{j_1} \cdots y_r^{j_r},$$

the sum running over all $n \in \mathbb{Z}_{\geq 1}^q$ and $j \in \mathbb{Z}_{\geq 0}^r$. Since the E_n are odd and the S_j are even, G_η is alternating in z and symmetric in y, and $\eta \mapsto G_\eta$ is a bijection from the graded dual onto $\mathcal{P}_{q,r}$.

The power sum $p_m(t) \in \mathcal{O}_k$ acts on $N_k^+ = \mathrm{Sym}^k \mathfrak{l}$ as the derivation induced by multiplication by z^m on $\mathfrak{l}$. Its transpose acts on the coefficients $\mathrm{Sym}(\mathfrak{l}^\vee)$ as the derivation $\sigma_n^a \mapsto \sigma_{n-m}^a$, where $\sigma_{n-m}^a = 0$ for $n < m$; this derivation commutes with $\mathfrak{l}$, and hence acts on $H^*(\mathfrak{l}, \mathfrak{sl}_2; \mathrm{Sym}(\mathfrak{l}^\vee))$ as an algebra derivation. On generating functions it multiplies $\sigma^a(z)$ by z^m, so it sends $\mathsf{S}_n \mapsto 2\,\mathsf{S}_{n-m}$ and $\mathsf{E}_n \mapsto \mathsf{E}_{n-m}$, where generators with indices out of range are zero. Dually, $p_m(t)$ multiplies G_η by $p_m(z) + 2p_m(y)$, which is the image of $p_m(t)$ under the ring map defining the $\mathcal{O}_k$-module structure of $\mathcal{P}_{q,r}$. As $p_1, \dots, p_k$ generate $\mathcal{O}_k$, the bijection is $\mathcal{O}_k$-linear.

Finally, the cup product is dual to Δ, and multiplication in $\mathrm{Sym}(\mathfrak{l}^\vee)$ is dual to the coproduct; so the map in the statement is the predual of the product $H^*(\mathfrak{l}, \mathfrak{sl}_2; \mathrm{Sym}^{k_1}(\mathfrak{l}^\vee)) \otimes H^*(\mathfrak{l}, \mathfrak{sl}_2; \mathrm{Sym}^{k_2}(\mathfrak{l}^\vee)) \to H^*(\mathfrak{l}, \mathfrak{sl}_2; \mathrm{Sym}^k(\mathfrak{l}^\vee))$. The product of two monomials is, after reordering the even factors, their concatenation; hence, up to the normalizations of the two dualities, its predual sends G_η to G_η, regarded as a polynomial in the two groups of variables. □

Proposition A.2. *For coprime monic polynomials f_1 and f_2, the map $\Delta_{\mathfrak{M}_{f_1}, \mathfrak{M}_{f_2}}$ is a quasi-isomorphism.*

Proof. By (A.1) we may assume that $k_i := \deg f_i \geq 1$; put $k = k_1 + k_2$ and $\mathcal{O}^\circ := \mathcal{O}^\circ_{k_1,k_2}$. Consider the $\mathcal{O}_k$-linear chain map

$$\Gamma := \Delta_{N_{k_1}^+, N_{k_2}^+} \circ \mathrm{CE}_\bullet(\iota_{k_1,k_2}) : \mathrm{CE}_\bullet(N_k^+) \longrightarrow \mathrm{CE}_\bullet(N_{k_1}^+) \otimes_{\mathbb{C}} \mathrm{CE}_\bullet(N_{k_2}^+),$$

where $\mathcal{O}_k$ acts on the target through $\mathcal{O}_{k_1} \otimes \mathcal{O}_{k_2}$, and its base change

$$\Gamma^\circ : \mathrm{CE}_\bullet(N_k^+) \otimes_{\mathcal{O}_k} \mathcal{O}^\circ \longrightarrow \big(\mathrm{CE}_\bullet(N_{k_1}^+) \otimes_{\mathbb{C}} \mathrm{CE}_\bullet(N_{k_2}^+)\big)\big[\mathrm{Res}^{-1}\big].$$

Since $N_k^+ \cong \mathcal{O}_k \otimes W_k^+$, every term $\mathrm{CE}_j(N_k^+) \cong \mathcal{O}_k \otimes (\Lambda^j(\mathfrak{l}/\mathfrak{sl}_2) \otimes W_k^+)_{\mathfrak{sl}_2}$ is a free $\mathcal{O}_k$-module, and similarly the terms of the target of Γ are free over $\mathcal{O}_{k_1} \otimes \mathcal{O}_{k_2}$. So both sides of Γ° are complexes of free $\mathcal{O}^\circ$-modules in degrees ≥ 0.

We claim that Γ° is a quasi-isomorphism. By flatness of $\mathcal{O}^\circ$ over $\mathcal{O}_k$, the Künneth formula over $\mathbb{C}$ and exactness of localization, the map induced by Γ° on homology is the base change of the map induced by Γ, which Lemma A.1 identifies, up to nonzero factors on the summands, with the direct sum of the maps

$$\mathcal{P}_{q,r} \otimes_{\mathcal{O}_k} \mathcal{O}^\circ \longrightarrow \bigoplus_{\substack{q_1+q_2=q,\ r_1+r_2=r \\ q_i+2r_i=k_i}} \big(\mathcal{P}_{q_1,r_1} \otimes_{\mathbb{C}} \mathcal{P}_{q_2,r_2}\big)\big[\mathrm{Res}^{-1}\big].$$

By the definition of the $\mathcal{O}_k$-module structure in (A.2), $\mathbb{C}[z,y]\otimes_{\mathcal{O}_k}\mathcal{O}^\circ$ is the algebra defined by the identity $f_1(u)f_2(u)=\prod_i(u-z_i)\prod_a(u-y_a)^2$. Apply Lemma 6.1 with weight one on the z_i and weight two on the y_a, and take invariants under $S_q\times S_r$, twisted by the sign character of S_q. The orbits of components are indexed by the pairs (q_1,r_1) above, and the stabilizer of a component permutes each of its four lists of roots separately. As in the proof of Lemma 6.2, projection onto one component in each orbit is therefore an isomorphism, and this projection is the displayed map. This proves the claim.

The cone of Γ° is thus an acyclic complex of free $\mathcal{O}^\circ$-modules in degrees ≥ 0. By induction on the degree, its modules of cycles are flat, so it remains acyclic after $\otimes_{\mathcal{O}^\circ}\mathbb{C}_{(f_1,f_2)}$, where $\mathbb{C}_{(f_1,f_2)}$ is the residue field of $\mathcal{O}^\circ$ at the point (f_1,f_2). Since $\mathrm{CE}_\bullet$ commutes with base change, the resulting map is $\Delta_{\mathfrak{M}_{f_1},\mathfrak{M}_{f_2}}\circ\mathrm{CE}_\bullet(\iota)$, where $\iota:\mathfrak{M}_{f_1f_2}\to\mathfrak{M}_{f_1}\otimes\mathfrak{M}_{f_2}$ is the isomorphism of Lemma 6.2. Hence $\Delta_{\mathfrak{M}_{f_1},\mathfrak{M}_{f_2}}$ is a quasi-isomorphism. □

This argument runs that of Proposition 6.7 in reverse: there the fibres are known and the finiteness of the BGG complexes passes the statement to the family, whereas here the family is known and flatness passes the statement to the fibres.

A.2. **Dévissage.**

Proof of Theorem 6.6. Write $M_1\perp M_2$ if Δ_{M_1,M_2} is a quasi-isomorphism. Since $\mathrm{CE}_\bullet$ and $\otimes_{\mathbb{C}}$ are exact and Δ is natural, the five lemma shows that, for fixed M_2, the class of modules M_1 with $M_1\perp M_2$ has the two-out-of-three property for short exact sequences; it is also closed under direct summands. The same holds in the second variable.

We first show: *if $Z\subset\mathbb{C}$ is finite and $\mathfrak{M}_f\perp M_2$ for every monic f with roots in Z, then $M\perp M_2$ for every $M\in\mathcal{L}(Z)$.* By the two-out-of-three property it suffices to show that $V_{2\mathbf{p}}(Z)\perp M_2$ for every $\mathbf{p}\in\mathbb{Z}^Z_{\geq 0}$, and we argue by induction on $|\mathbf{p}|$. If $\mathbf{p}=0$, then $V_{2\mathbf{p}}(Z)=\mathbb{C}=\mathfrak{M}_1$. Otherwise let $f=\prod_a(u-a)^{p_a}$. By Lemmas 6.2 and 6.3, $\mathfrak{M}_f\cong\bigotimes_a\mathfrak{M}_{(u-a)^{p_a}}$ surjects onto $\overline{\mathfrak{M}}_f:=\bigotimes_a\mathrm{Sym}^{p_a}(V_2)(a)$, and the kernel $\mathfrak{M}'_f$ is filtered by tensor products of subquotients of the filtrations $F^\bullet$, with at least one factor coming from some F^1. Since $\mathfrak{M}_{(u-a)^{p_a}}\cong V_2^{\otimes p_a}$ contains V_{2p_a} only once, in the quotient $\mathrm{Sym}^{p_a}(V_2)$, all constituents of F^1 are of the form $V_{2p'}(a)$ with $p'<p_a$. Hence $\mathfrak{M}'_f$ has a filtration with subquotients $V_{2\mathbf{p}'}(Z)$, $|\mathbf{p}'|<|\mathbf{p}|$, so that $\mathfrak{M}'_f\perp M_2$ by induction. Since $\mathfrak{M}_f\perp M_2$ by hypothesis, we get $\overline{\mathfrak{M}}_f\perp M_2$. Finally, $V_{2\mathbf{p}}(Z)$ is a direct summand of $\overline{\mathfrak{M}}_f$, because V_{2p_a} is a direct summand of $\mathrm{Sym}^{p_a}(V_2)$; hence $V_{2\mathbf{p}}(Z)\perp M_2$.

Now let f_2 be a monic polynomial with roots in Z_2. By (A.1) and Proposition A.2, $\mathfrak{M}_f\perp\mathfrak{M}_{f_2}$ for every monic f with roots in Z_1, so $M_1\perp\mathfrak{M}_{f_2}$ by the claim. Since f_2 was arbitrary, the claim with the roles of the two variables exchanged gives $M_1\perp M_2$. □

Remark A.3. For a single evaluation module $V_{2p}(a)$, the complex of Theorem 3.1 has the single nonzero term $(V_{2p}\otimes V_{2p})^{\mathfrak{sl}_2}[d_p]$, so $H_*(\mathfrak{l},\mathfrak{sl}_2;V_{2p}(a))$ is one-dimensional, in degree p. With Theorem 6.6 it follows that, for pairwise distinct $a_1,\dots,a_n$, the homology $H_*\big(\mathfrak{l},\mathfrak{sl}_2;\bigotimes_i V_{2p_i}(a_i)\big)$ is one-dimensional, in degree $\sum_i p_i$. Compare Teleman's Borel–Weil–Bott theory for current algebras [Tel95, Tel96, Tel98], which computes the cohomology of $\mathfrak{g}[\Sigma]$, for an affine curve Σ, with coefficients in the tensor product of an integrable representation and of evaluation modules. We do not use this.

References

[AB10] D. Arinkin and R. Bezrukavnikov, "Perverse coherent sheaves," Mosc. Math. J. **10**, no.1, 3–29 (2010) [arXiv:0902.0349 [math.AG]].

[AGO87] R. C. Arcuri, J. F. Gomes and D. I. Olive, "Conformal subalgebras and symmetric spaces," Nucl. Phys. B **285**, 327–339 (1987).

[BG25] P. H. Balduf and D. Gaiotto, "Combinatorial proof of a non-renormalization theorem," JHEP **05**, 120 (2025) doi:10.1007/JHEP05(2025)120 [arXiv:2408.03192 [math-ph]].

[BBBDN20] C. Beem, D. Ben-Zvi, M. Bullimore, T. Dimofte and A. Neitzke, "Secondary products in supersymmetric field theory," Annales Henri Poincaré **21**, no.4, 1235–1310 (2020) doi:10.1007/s00023-020-00888-3 [arXiv:1809.00009 [hep-th]].

[BD04] A. Beilinson and V. Drinfeld, *Chiral Algebras*, American Mathematical Society Colloquium Publications **51**, American Mathematical Society, Providence, RI (2004).

[BF08] R. Bezrukavnikov and M. Finkelberg, "Equivariant Satake category and Kostant–Whittaker reduction," Mosc. Math. J. **8**, no.1, 39–72 (2008) [arXiv:0707.3799 [math.RT]].

[BFM05] R. Bezrukavnikov, M. Finkelberg and I. Mirković, "Equivariant homology and K-theory of affine Grassmannians and Toda lattices," Compos. Math. **141**, no.3, 746–768 (2005) doi:10.1112/S0010437X04001228 [arXiv:math/0306413 [math.AG]].

[BFN18] A. Braverman, M. Finkelberg and H. Nakajima, "Towards a mathematical definition of Coulomb branches of 3-dimensional $\mathcal{N} = 4$ gauge theories, II," Adv. Theor. Math. Phys. **22**, no.5, 1071–1147 (2018) doi:10.4310/ATMP.2018.v22.n5.a1 [arXiv:1601.03586 [math.RT]].

[BH93] W. Bruns and J. Herzog, *Cohen–Macaulay Rings*, Cambridge Studies in Advanced Mathematics **39**, Cambridge University Press, Cambridge (1993).

[BE73] D. A. Buchsbaum and D. Eisenbud, "What makes a complex exact?," J. Algebra **25**, 259–268 (1973).

[CW19] S. Cautis and H. Williams, "Cluster theory of the coherent Satake category," J. Amer. Math. Soc. **32**, no.3, 709–778 (2019) [arXiv:1801.08111 [math.RT]].

[CW23] S. Cautis and H. Williams, "Canonical bases for Coulomb branches of 4d $\mathcal{N} = 2$ gauge theories," [arXiv:2306.03023 [math.AG]].

[Che55] C. Chevalley, "Invariants of finite groups generated by reflections," Amer. J. Math. **77**, 778–782 (1955).

[CGS16] C. Cordova, D. Gaiotto and S. H. Shao, "Infrared Computations of Defect Schur Indices," JHEP **11**, 106 (2016) doi:10.1007/JHEP11(2016)106 [arXiv:1606.08429 [hep-th]].

[ESW22] C. Elliott, P. Safronov and B. R. Williams, "A taxonomy of twists of supersymmetric Yang–Mills theory," Selecta Math. **28**, no.4, 73 (2022) doi:10.1007/s00029-022-00786-y [arXiv:2002.10517 [math-ph]].

[Fei88] B. L. Feigin, "On the cohomology of the Lie algebra of vector fields and of the current algebra," Selecta Math. Soviet. **7**, 49–62 (1988).

[Fre81] I. B. Frenkel, "Two constructions of affine Lie algebra representations and boson–fermion correspondence in quantum field theory," J. Funct. Anal. **44**, 259–327 (1981).

[FGT08] S. Fishel, I. Grojnowski and C. Teleman, "The strong Macdonald conjecture and Hodge theory on the loop Grassmannian," Ann. of Math. (2) **168**, no.1, 175–220 (2008) doi:10.4007/annals.2008.168.175 [arXiv:math/0411355 [math.AG]].

[GL76] H. Garland and J. Lepowsky, "Lie algebra homology and the Macdonald–Kac formulas," Invent. Math. **34**, 37–76 (1976).

[Han90] P. Hanlon, "Some conjectures and results concerning the homology of nilpotent Lie algebras," Adv. Math. **84**, 91–134 (1990).

[HS53] G. Hochschild and J.-P. Serre, "Cohomology of Lie algebras," Ann. of Math. (2) **57**, 591–603 (1953).

[Kap06] A. Kapustin, "Holomorphic reduction of N=2 gauge theories, Wilson-'t Hooft operators, and S-duality," [arXiv:hep-th/0612119 [hep-th]].

[KS09] A. Kapustin and N. Saulina, "The algebra of Wilson–'t Hooft operators," Nucl. Phys. B **814**, 327–365 (2009) doi:10.1016/j.nuclphysb.2009.02.004 [arXiv:0710.2097 [hep-th]].

[K26] A. Z. Khan, "Poisson Vertex Algebra of Seiberg–Witten Theory," [arXiv:2604.03500 [hep-th]].

[Kos61] B. Kostant, "Lie algebra cohomology and the generalized Borel–Weil theorem," Ann. of Math. (2) **74**, 329–387 (1961).

[Kum02] S. Kumar, *Kac–Moody Groups, their Flag Varieties and Representation Theory*, Progress in Mathematics **204**, Birkhäuser, Boston (2002).

[Lep79] J. Lepowsky, "Generalized Verma modules, loop space cohomology and MacDonald-type identities," Ann. Sci. École Norm. Sup. (4) **12**, no.2, 169–234 (1979) doi:10.24033/asens.1365

[Mac95] I. G. Macdonald, *Symmetric Functions and Hall Polynomials*, 2nd ed., Oxford Mathematical Monographs, Clarendon Press, Oxford (1995).

[Mat86] H. Matsumura, *Commutative Ring Theory*, Cambridge Studies in Advanced Mathematics **8**, Cambridge University Press, Cambridge (1986).

[MV07] I. Mirković and K. Vilonen, "Geometric Langlands duality and representations of algebraic groups over commutative rings," Ann. of Math. (2) **166**, no.1, 95–143 (2007) doi:10.4007/annals.2007.166.95 [arXiv:math/0401222 [math.RT]].

[Niu22] W. Niu, "Local operators of 4d $\mathcal{N} = 2$ gauge theories from the affine Grassmannian," Adv. Theor. Math. Phys. **26**, no.9, 3207–3247 (2022) doi:10.4310/ATMP.2022.v26.n9.a10 [arXiv:2112.12164 [math.AG]].

[OY20] J. Oh and J. Yagi, "Poisson vertex algebras in supersymmetric field theories," Lett. Math. Phys. **110**, no.8, 2245–2275 (2020) doi:10.1007/s11005-020-01290-0 [arXiv:1908.05791 [hep-th]].

[PS73] C. Peskine and L. Szpiro, "Dimension projective finie et cohomologie locale," Publ. Math. IHÉS **42**, 47–119 (1973) doi:10.1007/BF02685877.

[SW94] N. Seiberg and E. Witten, "Electric-magnetic duality, monopole condensation, and confinement in N=2 supersymmetric Yang–Mills theory," Nucl. Phys. B **426**, 19–52 (1994) [erratum: Nucl. Phys. B **430**, 485–486 (1994)] doi:10.1016/0550-3213(94)90124-4 [arXiv:hep-th/9407087 [hep-th]].

[Stacks] The Stacks Project Authors, *The Stacks Project*, https://stacks.math.columbia.edu (2026).

[Tak71] S. J. Takiff, "Rings of invariant polynomials for a class of Lie algebras," Trans. Amer. Math. Soc. **160**, 249–262 (1971) doi:10.1090/S0002-9947-1971-0281839-9.

[Tel95] C. Teleman, "Lie algebra cohomology and the fusion rules," Commun. Math. Phys. **173**, 265–311 (1995).

[Tel96] C. Teleman, "Verlinde factorization and Lie algebra cohomology," Invent. Math. **126**, 249–263 (1996).

[Tel98] C. Teleman, "Borel–Weil–Bott theory on the moduli stack of G-bundles over a curve," Invent. Math. **134**, 1–57 (1998).